

ADR Dub Studio

Dubbing and ADR studio for macOS and Windows

User manual · version 2.22.0

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1. Introduction

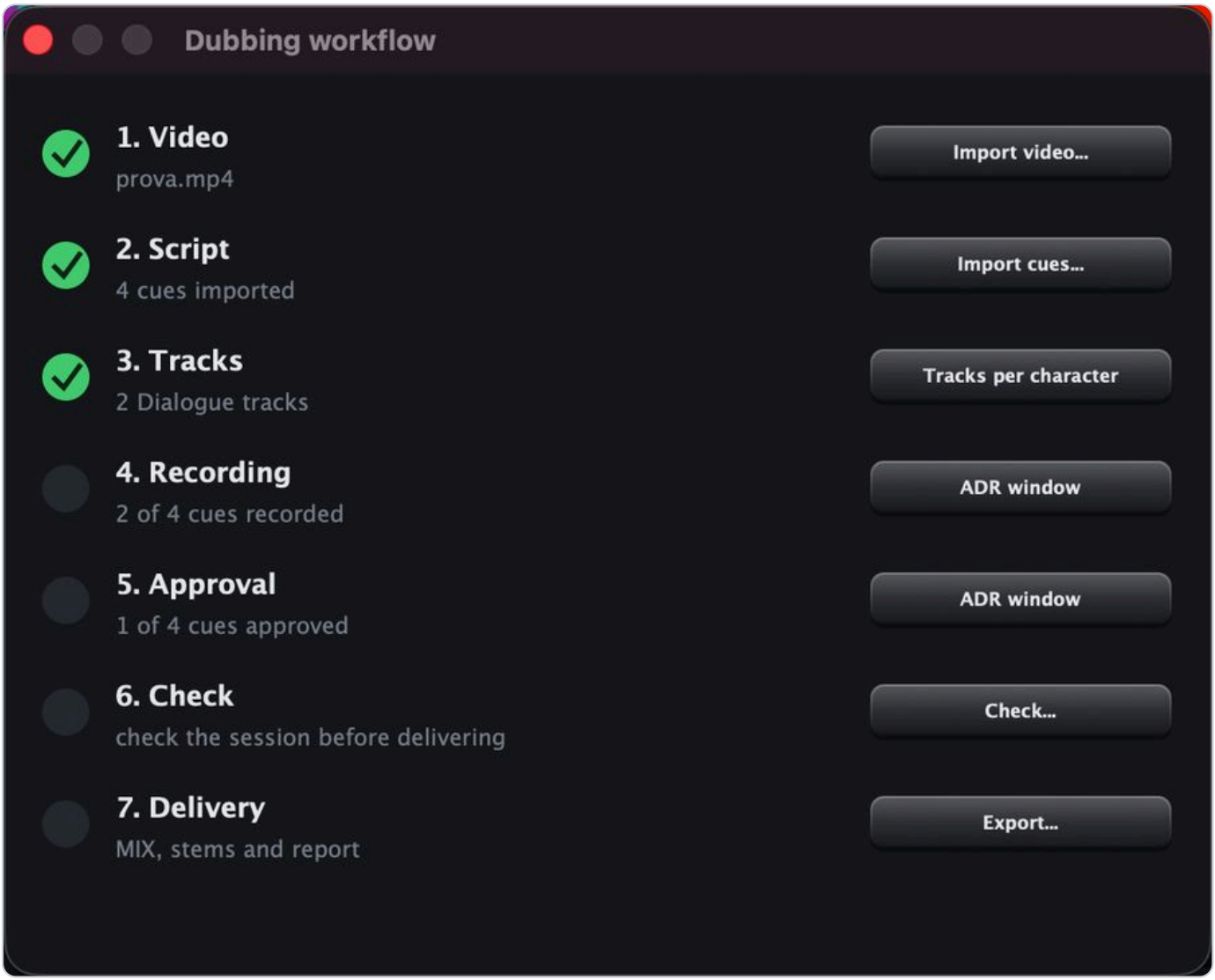
ADR Dub Studio is a compact digital studio designed for **dubbing** and **automated dialogue replacement (ADR)**: you import a video, import or write the list of lines (the *cues*), record several *takes* of every line right on the picture, pick the best ones, align them to the guide, mix, and deliver the material (mix and stems) with a check report. The program runs on macOS (Apple Silicon and Intel) and on Windows, reads and writes WAV, AIFF and FLAC files, and can host VST3 plug-ins as inserts.

This manual describes every button and every function. The pictures are real screenshots of the program: the **orange numbers** refer to the lists below each picture. In the text, keyboard shortcuts are written with **Cmd** for macOS; on Windows use **Ctrl** instead, and **Alt** corresponds to **Option**.

1.1 The workflow at a glance

1. **Video** – import the film: its original audio becomes the *Guide* track and the picture appears at the top right, in sync with the timeline.
2. **Script** – import the cues from Pro Tools, from an EDL, from a CSV/TSV table or from SRT subtitles, or create them by hand; each cue has character, scene, in, out and dialogue.
3. **Tracks** – one track per character (the dialogue), plus M&E and effects; every role has its own bus and a ready-made chain of EQ, compressor and limiter.
4. **Recording** – the program counts down with beeps and a streamer over the picture, records between the cue's in and out points and creates a new take; you can repeat as often as you like.
5. **Choice and sync** – rate the takes, keep the best one and align it to the guide (even phrase by phrase with the Time Warp).
6. **Mix** – faders, pan, EQ, compressor, automatic ducking of the background under the dialogue, loudness measurement.
7. **Delivery** – mix and Dialogue / M&E / Effects stems with an HTML report, presets for TV and streaming, a review video with the mix.

> **Tip** Help > **Dubbing workflow** opens a checklist with these steps: each one ticks itself when done and has a button that launches the right command.



The workflow checklist (Help > Dubbing workflow).

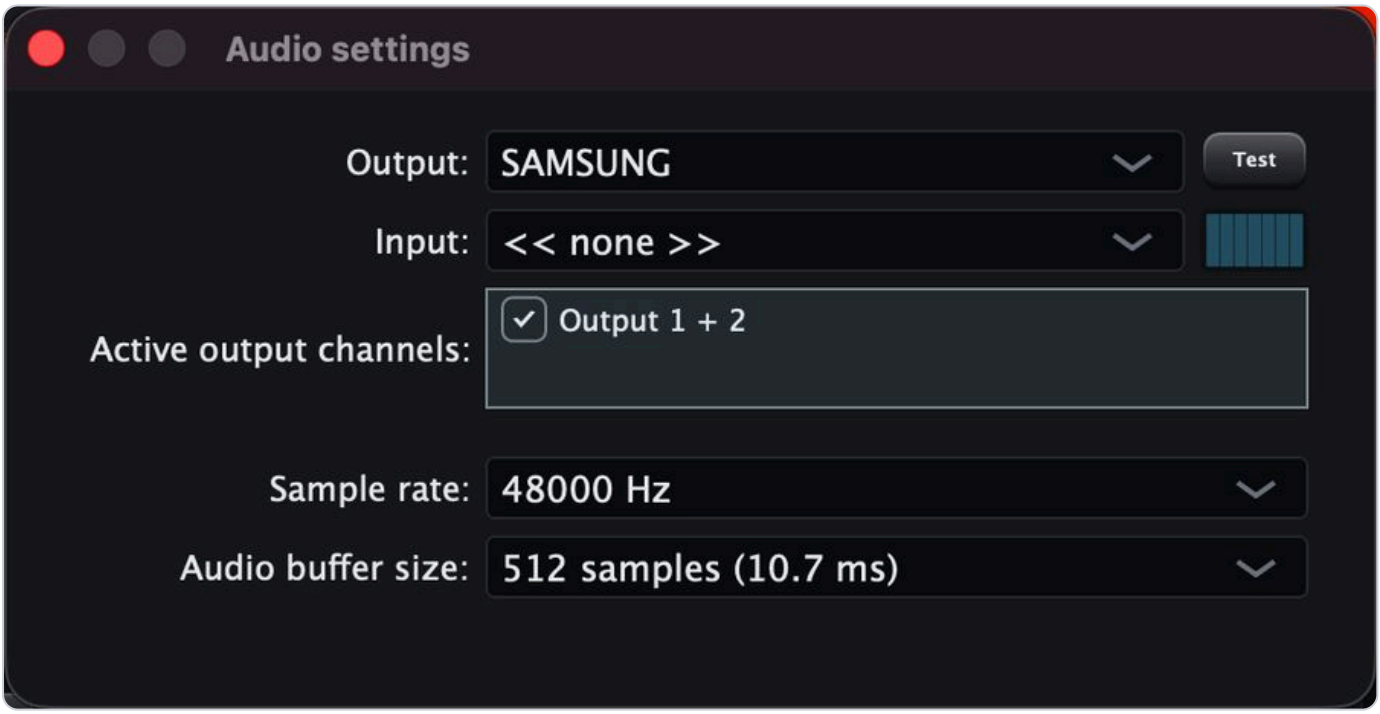
2. Installation and first launch

2.1 Installing the program

- **macOS**: download `ADR-Dub-Studio-…-macOS-universal.zip`, unzip it and drag **ADR Dub Studio** to the Applications folder. It is a universal build (Apple Silicon and Intel) and needs macOS 11 or later. On first launch, if macOS warns about an unidentified developer, open the app with right-click > Open.
- **Windows**: use the installer `…-Windows-x64-Setup.exe`, or unzip the portable version `…-Windows-x64-portable.zip` wherever you like. Exporting the video with the mix and some video features need `ffmpeg` (next to the program or in the PATH).
- **Microphone**: on macOS the system asks for permission the first time you record. If you deny it, you can re-enable it in System Settings > Privacy & Security > Microphone.

2.2 Choosing the audio interface

The program plays through the system's default audio device. To change it, or to choose the microphone input, open **File > Audio settings**.



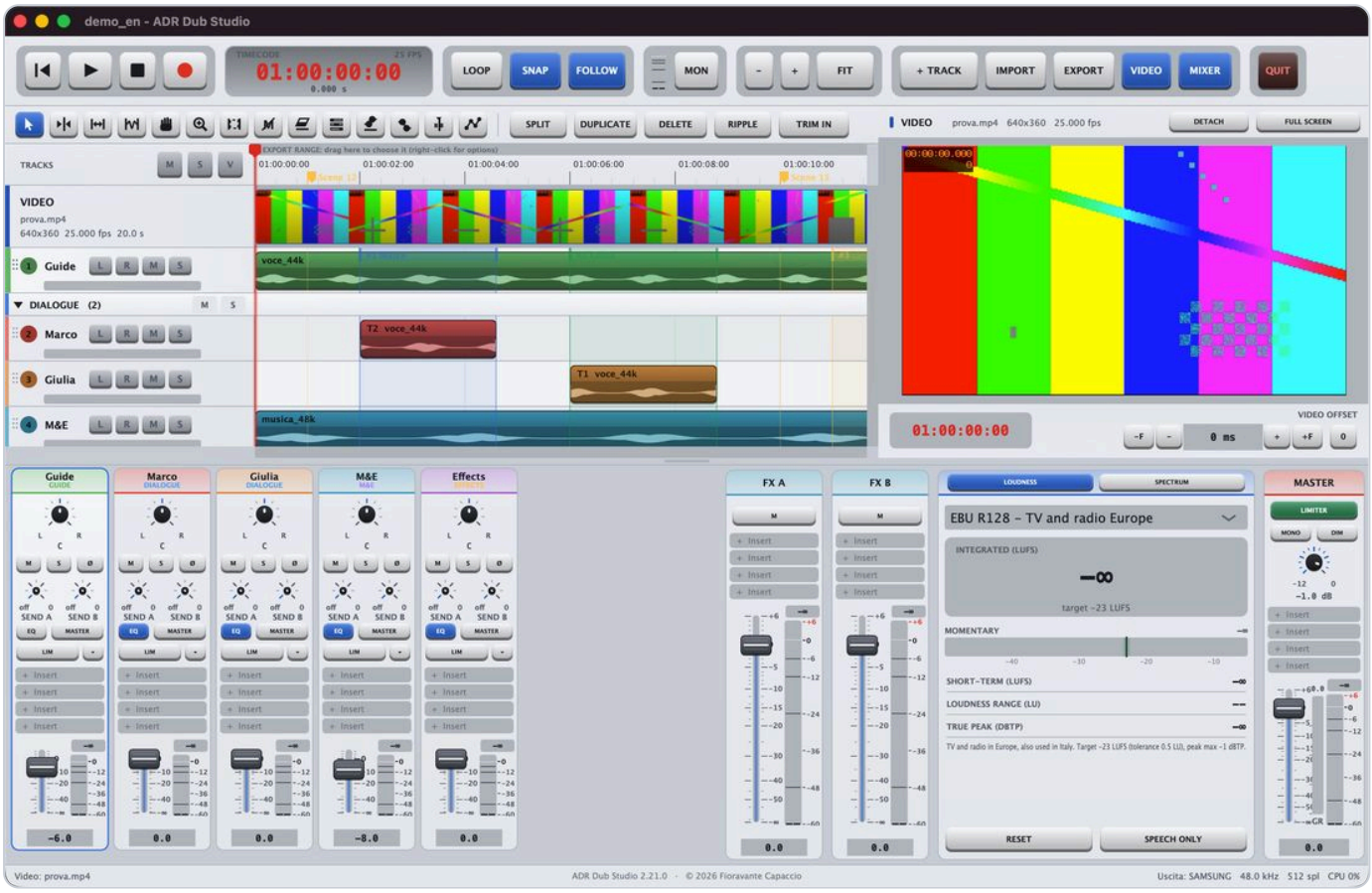
File > Audio settings: output, input, active channels, sample rate and buffer size.

- **Output / Input**: the available devices. At startup the program opens no input (this avoids stalls when the default input is a virtual device); the input is opened when needed for recording, or when you choose it here.
- **Active channels**: which outputs to use. With an interface of 4 or more outputs the second pair can feed the actor's headphones (see *Recording room*).
- **Sample rate / Buffer**: a smaller buffer lowers latency but loads the processor more. The status bar at the bottom right shows output, rate, buffer and CPU load.

- **Test**: plays a test tone on the chosen output.

2.3 Language and theme

The interface language (Italian or English) is changed from **Help > Lingua / Language** and applies on restart. The graphic theme is chosen in **View > Theme**: *Dark*, *Light*, *High contrast* and *Default 5* (neutral grey with teal accents). View > Layout sets the interface density (Compact, Standard, Large).



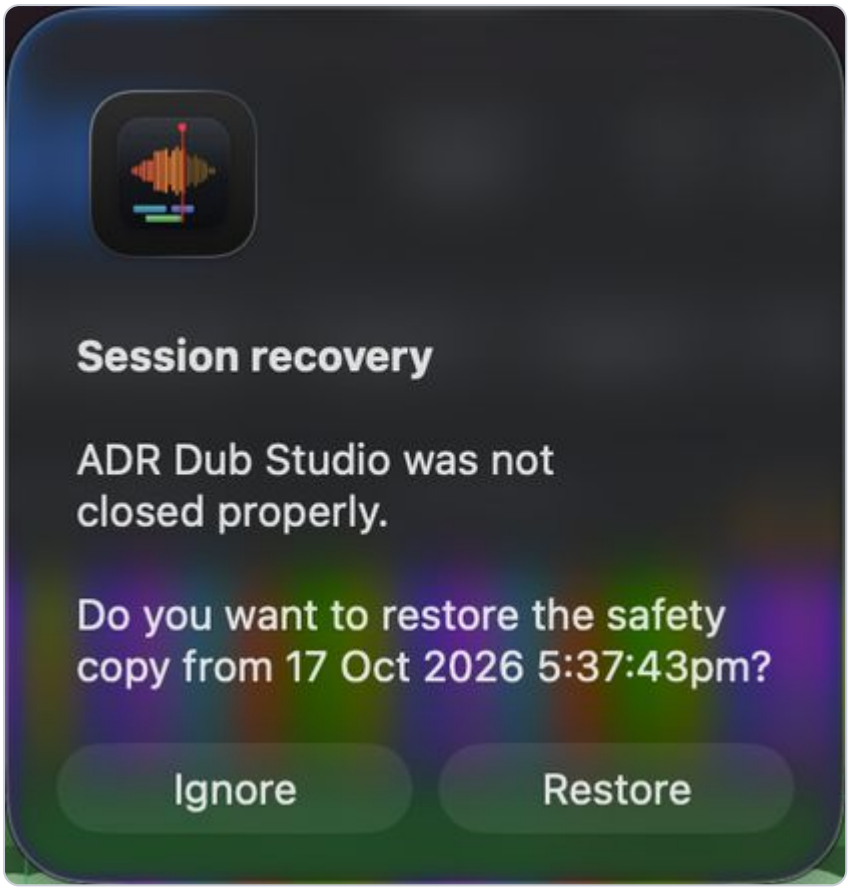
The Light theme.



The Default 5 theme.

2.4 Recovery after an abnormal exit

The program saves a safety copy of the session every few minutes. If it exits abnormally (crash, shutdown, power cut) the next launch offers to **restore** the safety copy or to **ignore** it.



The recovery request at launch.

3. The main window

The main window gathers the transport bar, the editing tools, the timeline with the tracks, the video viewer and the mixer. The parts can be shown or hidden with the **VIDEO** and **MIXER** buttons in the top bar.



The main window with a sample session.

1. Transport bar: rewind, play, stop, record and the timecode display.
2. Options: LOOP, SNAP, FOLLOW, input monitor, zoom and fit.
3. Session buttons: + TRACK, IMPORT, EXPORT, VIDEO, MIXER.
4. QUIT: closes the program (asks whether to save).
5. Editing tools (select, razor, slip, stretch, hand, zoom, range, mute, eraser, comp, Time Warp, envelope, scrub, automation).
6. Edit buttons: SPLIT, DUPLICATE, DELETE, RIPPLE, TRIM IN/OUT, mode.
7. Ruler with timecode, export range bar and markers.
8. Video lane with frame thumbnails.
9. Track and group headers.
10. Clips on the tracks and ADR cue regions (coloured zones).
11. Video viewer with ADR overlay and offset controls.
12. Mixer channels: one per track.

13. FX returns A and B.
14. Analysis panel: loudness and spectrum.
15. Master channel with limiter, MONO and DIM.
16. Status bar: version, audio output, rate, buffer and CPU.

3.1 The transport bar



The top bar (left part).

1. **Rewind** (Home): moves the playhead to the start.
2. **Play / Stop** (space bar).
3. **Stop**: stops playback; the playhead stays where it is.
4. **Record** (R): records on the armed tracks; if none is armed, on the selected track.
5. **Timecode** of the playhead and seconds; the format (23.976…60 fps) is set in Session settings.
6. **LOOP** (L): repeats the loop region (Shift + drag on the ruler to define it).
7. **SNAP** (N): edges of clips and cues snap to clips, playhead, markers and frames.
8. **FOLLOW** (F): the view follows the playhead during playback.
9. **Input level** meter of the microphone.
10. **MON** (I): hear the input while recording (use headphones to avoid feedback).
11. **–** and **+**: horizontal zoom of the timeline.
12. **FIT**: shows the whole project in the window (Shift + Z).



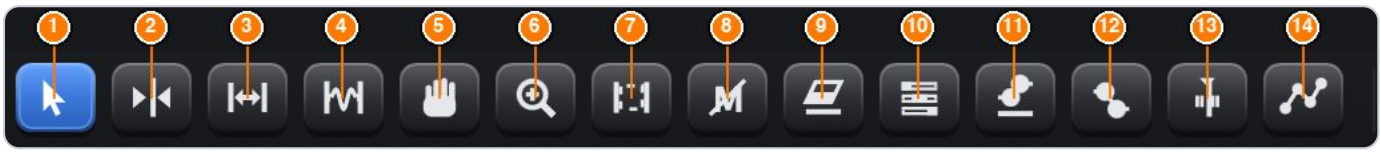
The top bar (right part).

1. **+ TRACK**: adds a track (Shift + Cmd + T).
2. **IMPORT**: imports audio files (Cmd + I).
3. **EXPORT**: exports the mix (Cmd + E).
4. **VIDEO**: shows or hides the video viewer (Shift + V).
5. **MIXER**: shows or hides the mixer (Shift + M).

6 **QUIT**: closes the program after asking whether to save.

Playback stops by itself one second after the end of the content (clips, video or cues). At the end of a recording, or with **Transport > On stop the playhead returns to where it started**, the playhead goes back to where it began.

3.2 The tools and edit buttons



The tools.

1 **Select** (V)
2 **Razor** (B)
3 **Slip** (Y)
4 **Stretch** (E)
5 **Hand** (H)
6 **Zoom** (Z)
7 **Range** (X)
8 **Mute** (K)
9 **Eraser** (G)
10 **Comp** (C)
11 **Time Warp** (O)
12 **Clip envelope** (D)
13 **Scrub** (J)
14 **Automation** (U)



The edit buttons.

1 **SPLIT** (S)
2 **DUPLICATE** (Cmd + D)
3 **DELETE** (Del)
4 **RIPPLE** (Shift + Del)

5 **TRIM IN** (Q) – and, further right, TRIM OUT (W) and the edit mode button (NORMAL / RIPPLE / SHUFFLE).

Hovering over a button shows a tooltip that explains what it does. The tools are described in detail in the chapter *Clips and editing*.

3.3 The timeline



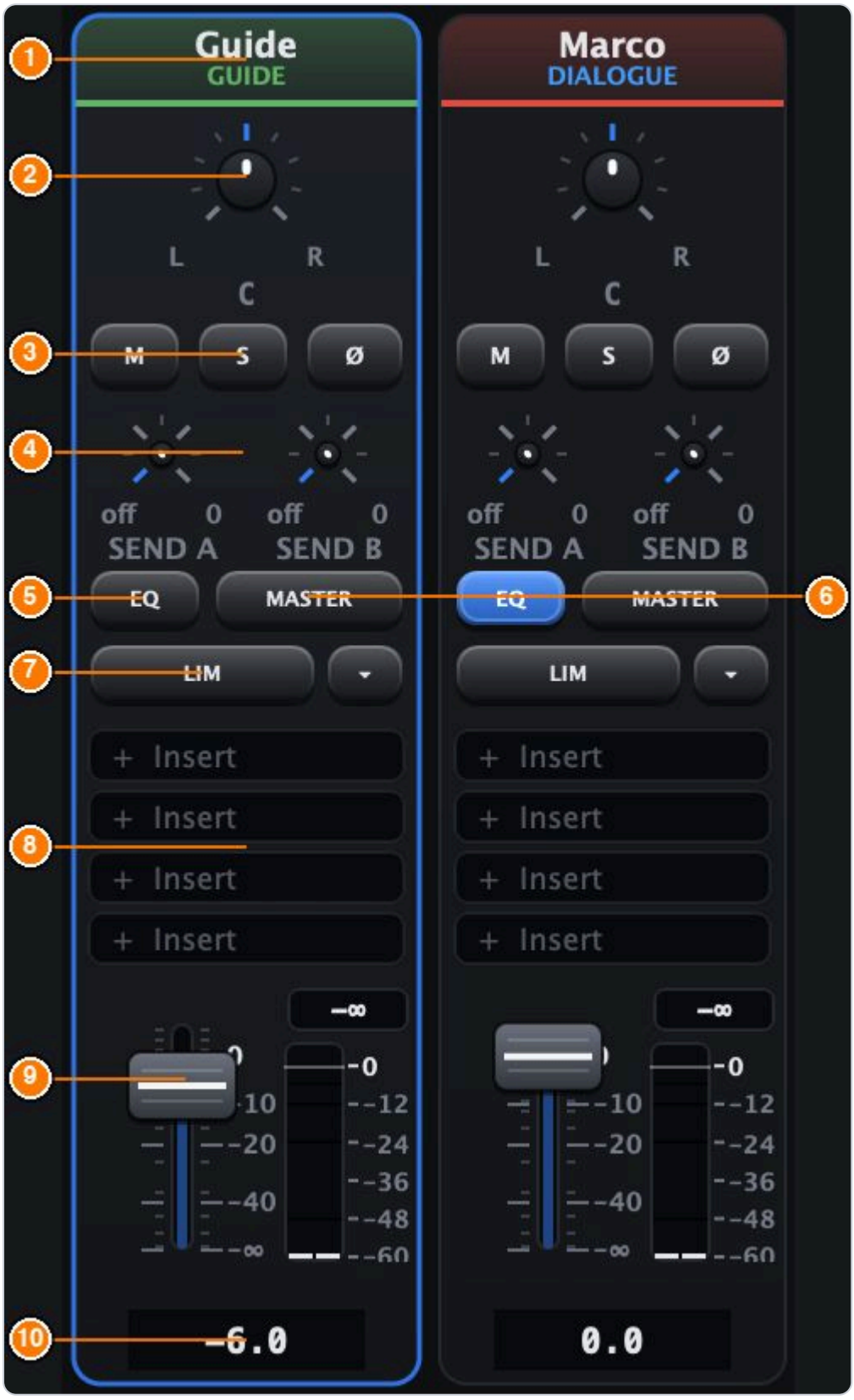
The timeline: ruler, video, track headers and clips.

1. Top-left corner: **M** clears all mutes, **S** clears all solos, **V** manages track visibility. Right-click chooses which controls the headers show.
2. **Export range** bar: drag to choose it, double-click to remove it, right-click for options.
3. **Ruler**: click to move the playhead, Shift + drag for the loop region, double-click to clear it; Alt + drag to scrub.
4. **Markers** (here "Scene 12"): click to jump, double-click to rename, drag to move, right-click for the menu.
5. **Video lane**: drag it to change the video offset against the audio.
6. Number (track colour) and **name**: double-click to rename, drag the handle on the left to reorder the track.
7. **L** locks, **R** arms for recording, **M** mutes, **S** solos.
8. Track level meter.
9. Collapsible **track group** (here "Dialogue") with M and S for all its tracks.
10. **Clip** (here an ADR take: "T2" is the second take; inactive takes are greyed out).
11. **ADR cue region**: the coloured zone in the background, with handles on its edges, shows the cue's in and out points.

- **Moving the view**: wheel or two-finger drag to scroll; **Cmd + wheel** (or trackpad pinch) to zoom horizontally; **Alt + wheel** for track height; the *Hand* tool to drag the view.
- **Track height**: drag the bottom edge of a track (Shift: all); double-click the edge for the default height; **View > Fit track height** makes all of them fit the window.
- **Selection**: click a clip; Shift + click adds; drag over an empty space for a selection rectangle; Cmd + A selects everything.
- **Context menus**: right-click a clip, a track, the ruler or a marker.

3.4 The mixer

The mixer shows one channel for every track, the two FX returns, the analysis panel and the master. At the top each channel shows the name and, below it, the track **role** (Guide, Dialogue, M&E, Effects).



A mixer channel (Guide track).

1. Track name and **role**.

2. **Pan** knob from left (L) to right (R); double-click returns to the centre (C).
3. **M** (mute), **S** (solo) and **Ø**: inverts the polarity (phase).
4. **SEND A** and **SEND B**: sends to the FX returns, post-fader; double-click turns them off.
5. **EQ**: opens the EQ and compressor editor; it is lit (blue) when the track uses one.
6. Track **output**: MASTER or a group bus (Dialogue, M&E, Effects, Guide).
7. **LIM**: track limiter after the fader (the arrow next to it sets the ceiling).
8. **Inserts** (4 per channel): click "+ Insert" to add a built-in effect or a VST3 plug-in; right-click to move, bypass, open, remove.
9. Volume **fader** (in dB; double-click = 0 dB) and peak meter with scale.
10. Fader value box: you can type a value.

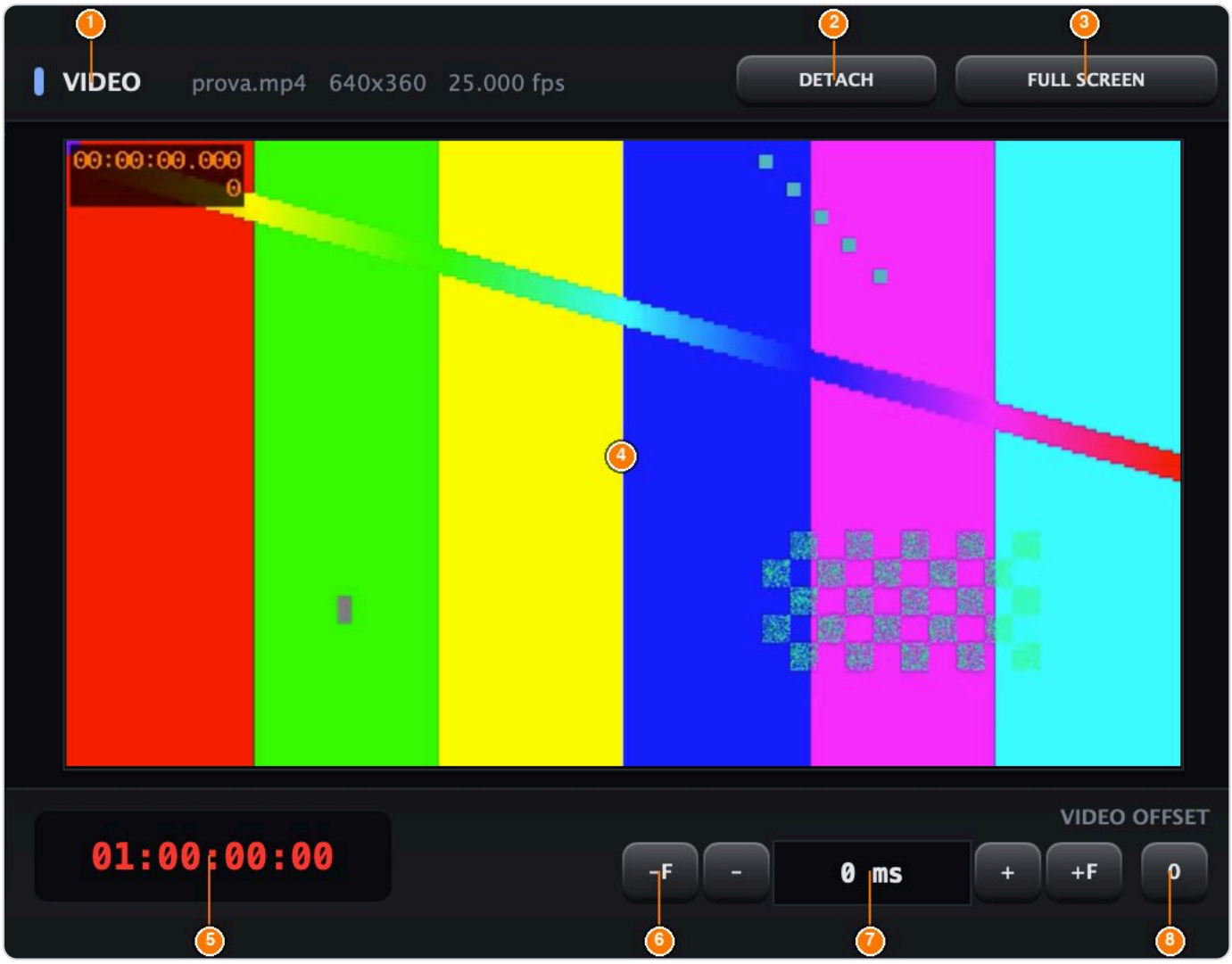


FX returns, analysis panel (loudness) and master.

1. **FX A / FX B**: effect returns with four inserts, fader and mute (M); they receive the tracks' sends.
2. **LOUDNESS / SPECTRUM** tabs of the analysis panel.
3. **Loudness standard** (EBU R128, ATSC A/85, ARIB, Netflix, podcast, streaming, audiobooks): sets target and ceiling.
4. **Integrated** loudness (LUFS) with the chosen programme target.
5. **Momentary** loudness and, below, short-term, loudness range and true peak.
6. **RESET**: resets the measurements.
7. **SPEECH ONLY**: counts only the blocks that contain speech ("dialogue-gated" measurement).

8. Master **LIMITER** (3 ms look-ahead, 80 ms release): no peak exceeds the ceiling.
9. **MONO** (compatibility check) and **DIM** (-20 dB): listening only, they do not affect export and measurements.
10. **Limiter ceiling** knob (double-click: -1 dB).
11. Master fader and meter.

3.5 The video viewer



The video panel.

1. Video name and format.
2. **DETACH**: opens the video in a separate window (even on another screen); again to re-attach.
3. **FULL SCREEN**: the video full screen (Esc to exit).
4. **Picture**, with the frame timecode and, during ADR cues, the overlay.
5. Playhead timecode.
6. **–F / +F**: shift the video by one frame; **– / +**: by 1 ms (hold to repeat).
7. **Video offset** in milliseconds: positive = video late against the audio. Double-click to type a value.

8 **0**: resets the offset.

The video is read with the system engine (macOS) or with ffmpeg (Windows). The video's sound is extracted and put on the **Guide** track; the video can also be removed with **File > Remove video**.

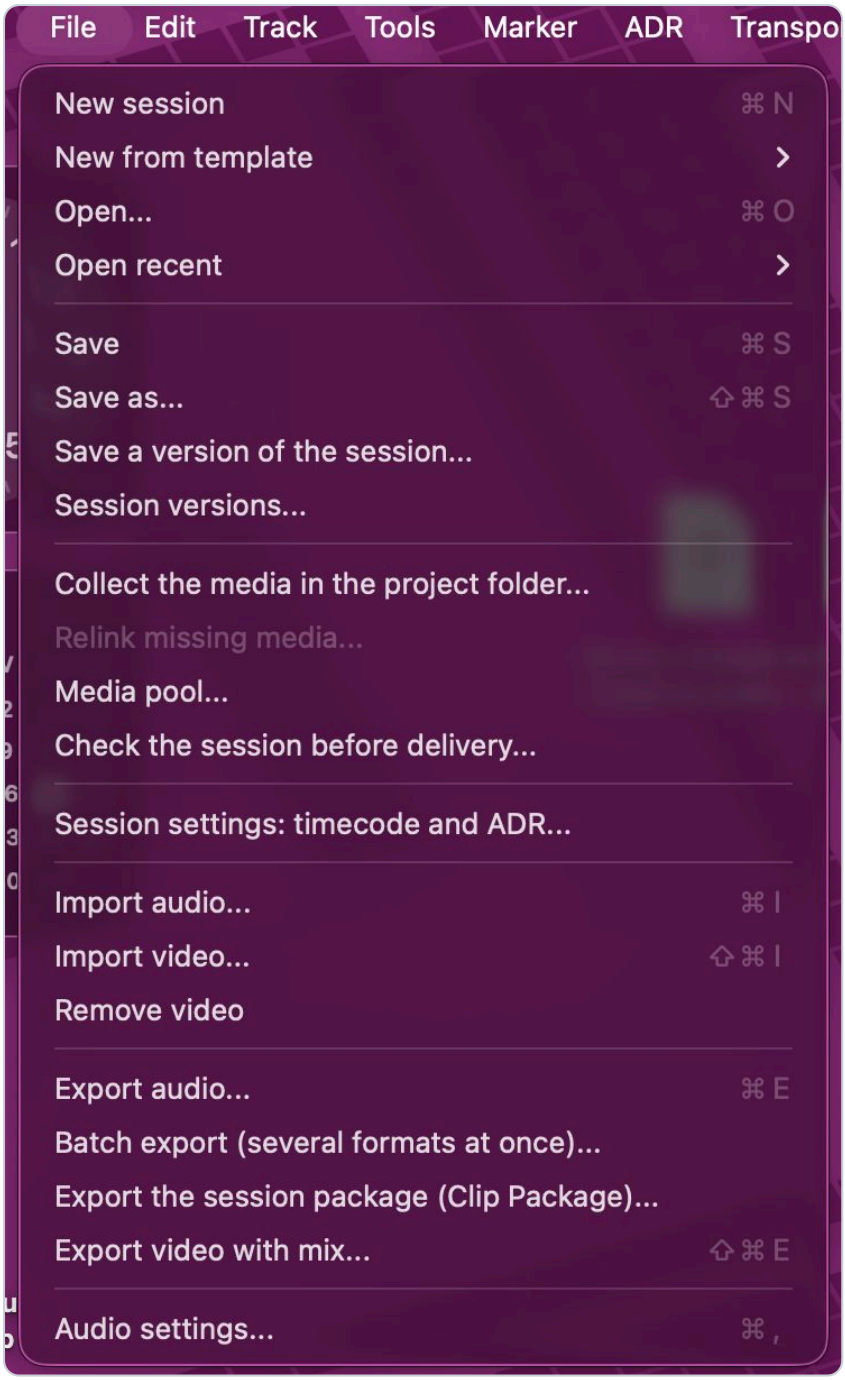


With an ADR cue selected, the streamer bar and the line of dialogue appear over the picture.

> **Note** When a cue is recorded the video shows itself if it was hidden. The overlay (cue number, character, scene, timecode, line) is chosen in *Session settings*.

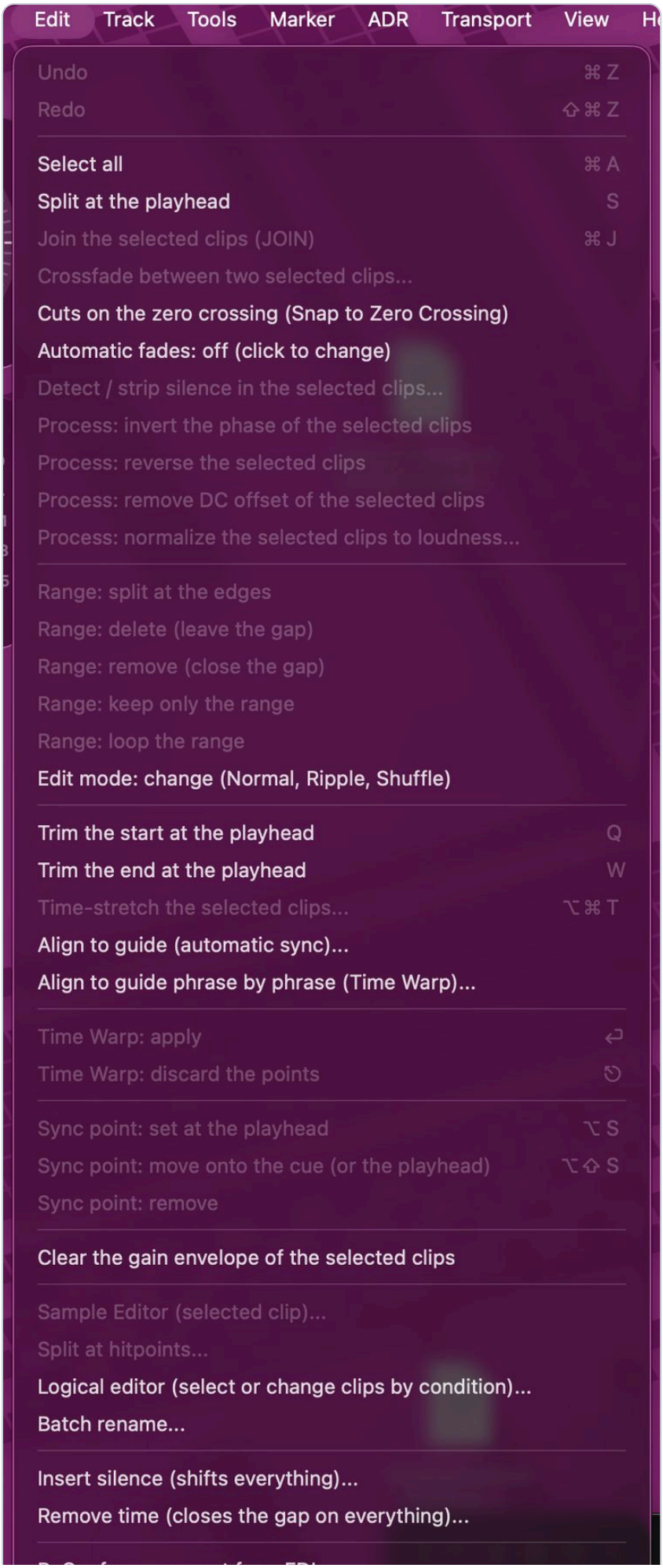
3.6 The menus

All functions can be reached from the menus at the top. Below, each menu with its shortcuts; the functions are explained in the following chapters.



The File menu

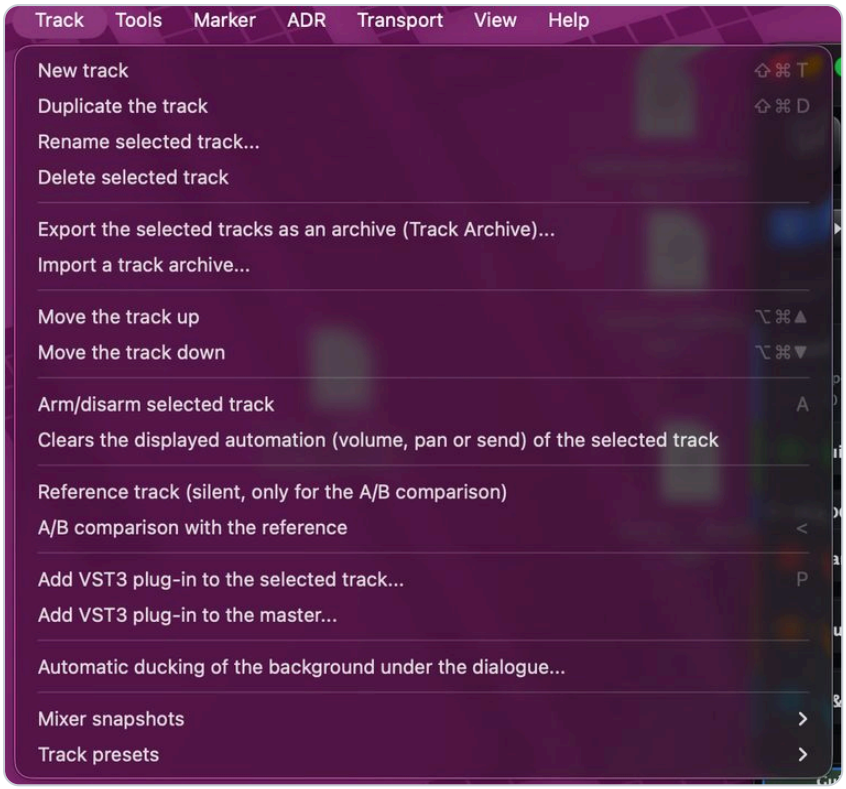
Command	Shortcut
New session	`Cmd/Ctrl+N`
Open...	`Cmd/Ctrl+O`
Save	`Cmd/Ctrl+S`
Save as...	`Cmd/Ctrl+Shift+S`
Import audio...	`Cmd/Ctrl+I`
Export audio...	`Cmd/Ctrl+E`
Audio settings...	`Cmd/Ctrl+,`
Import video...	`Cmd/Ctrl+Shift+I`
Export video with mix...	`Cmd/Ctrl+Shift+E`
Quit	`Cmd/Ctrl+Q`


Edit
Track
Tools
Marker
ADR
Transport
View
Undo ⌘ Z
Redo ⇧ ⌘ Z
Select all ⌘ A
Split at the playhead S
Join the selected clips (JOIN) ⌘ J
Crossfade between two selected clips...
Cuts on the zero crossing (Snap to Zero Crossing)
Automatic fades: off (click to change)
Detect / strip silence in the selected clips...
Process: invert the phase of the selected clips
Process: reverse the selected clips
Process: remove DC offset of the selected clips
Process: normalize the selected clips to loudness...
Range: split at the edges
Range: delete (leave the gap)
Range: remove (close the gap)
Range: keep only the range
Range: loop the range
Edit mode: change (Normal, Ripple, Shuffle)
Trim the start at the playhead Q
Trim the end at the playhead W
Time-stretch the selected clips... ⌥ ⌘ T
Align to guide (automatic sync)...
Align to guide phrase by phrase (Time Warp)...
Time Warp: apply
Time Warp: discard the points
Sync point: set at the playhead ⌥ S
Sync point: move onto the cue (or the playhead) ⌥ ⇧ S
Sync point: remove
Clear the gain envelope of the selected clips
Sample Editor (selected clip)...
Split at hitpoints...
Logical editor (select or change clips by condition)...
Batch rename...
Insert silence (shifts everything)...
Remove time (closes the gap on everything)...



The Edit menu

Command	Shortcut
Undo	`Cmd/Ctrl+Z`
Redo	`Cmd/Ctrl+Shift+Z`
Select all	`Cmd/Ctrl+A`
Duplicate	`Cmd/Ctrl+D`
Delete	`Delete` / `Backspace`
Split at the playhead	`S`
Join the selected clips (JOIN)	`Cmd/Ctrl+J`
Time-stretch the selected clips...	`Cmd/Ctrl+Alt+T`
Time Warp: apply	`Return`
Time Warp: discard the points	`Esc`
Sync point: set at the playhead	`Alt+S`
Sync point: move onto the cue (or the playhead)	`Shift+Alt+S`
Ripple delete	`Shift+Delete` / `Shift+Backspace`
Trim the start at the playhead	`Q`
Trim the end at the playhead	`W`



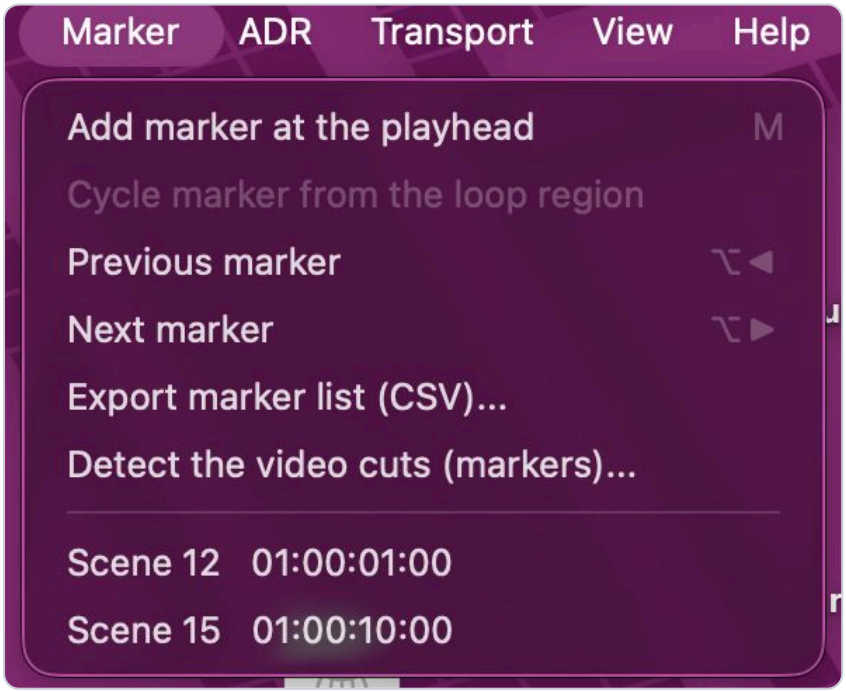
The Track menu

Command	Shortcut
New track	`Cmd/Ctrl+Shift+T`
Arm/disarm selected track	`A`
A/B comparison with the reference	`` ` ``
Move the track up	`Cmd/Ctrl+Alt+↑`
Duplicate the track	`Cmd/Ctrl+Shift+D`
Add VST3 plug-in to the selected track...	`P`



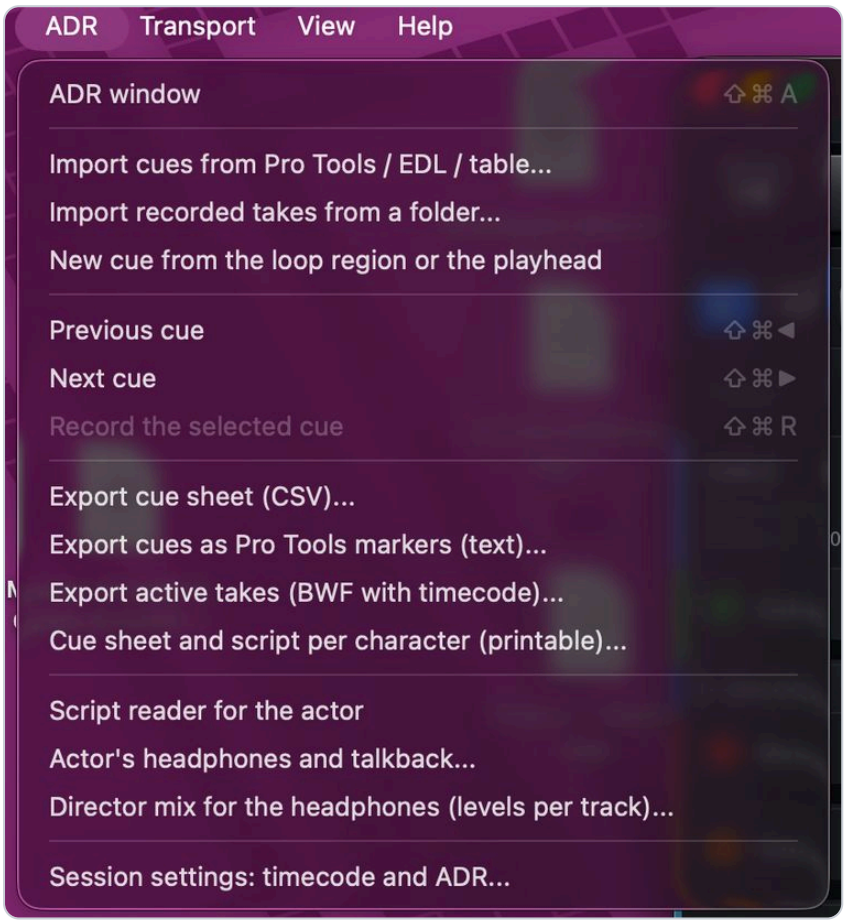
The Tools menu

Command	Shortcut
Selection	`V`
Razor	`B`
Stretch	`E`
Slip	`Y`
Range	`X`
Mute	`K`
Eraser	`G`
Comp	`C`
Hand	`H`
Zoom	`Z`
Time Warp	`0`
Clip envelope	`D`
Scrub	`J`
Automation	`U`



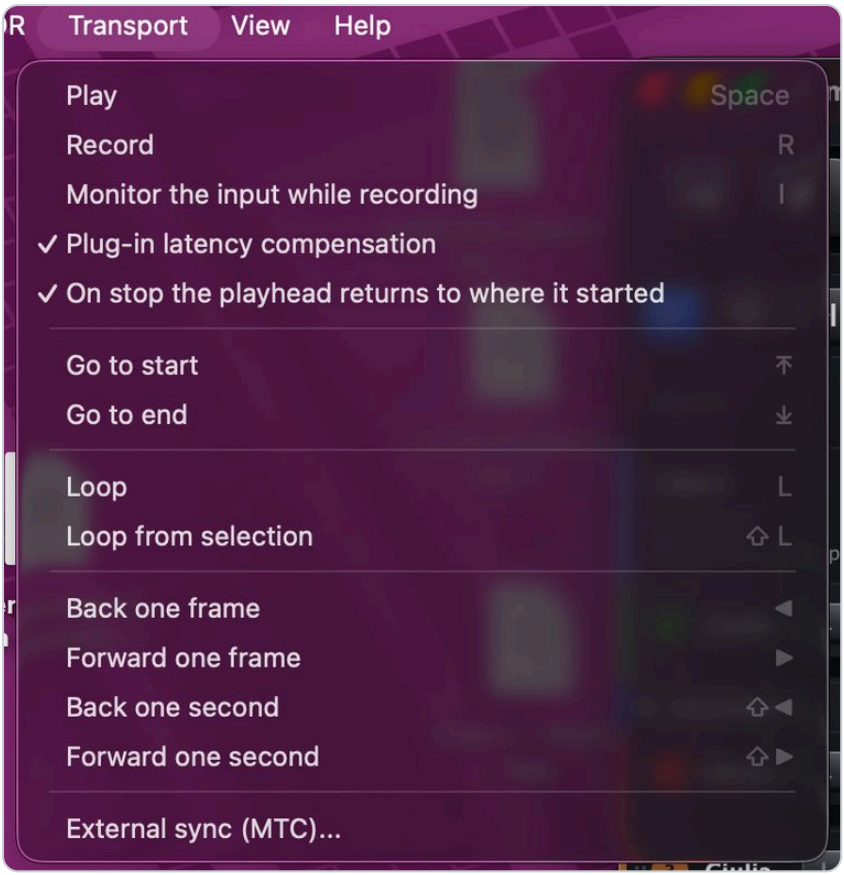
The Marker menu

Command	Shortcut
Add marker at the playhead	`M`
Previous marker	`Alt+←`
Next marker	`Alt+→`



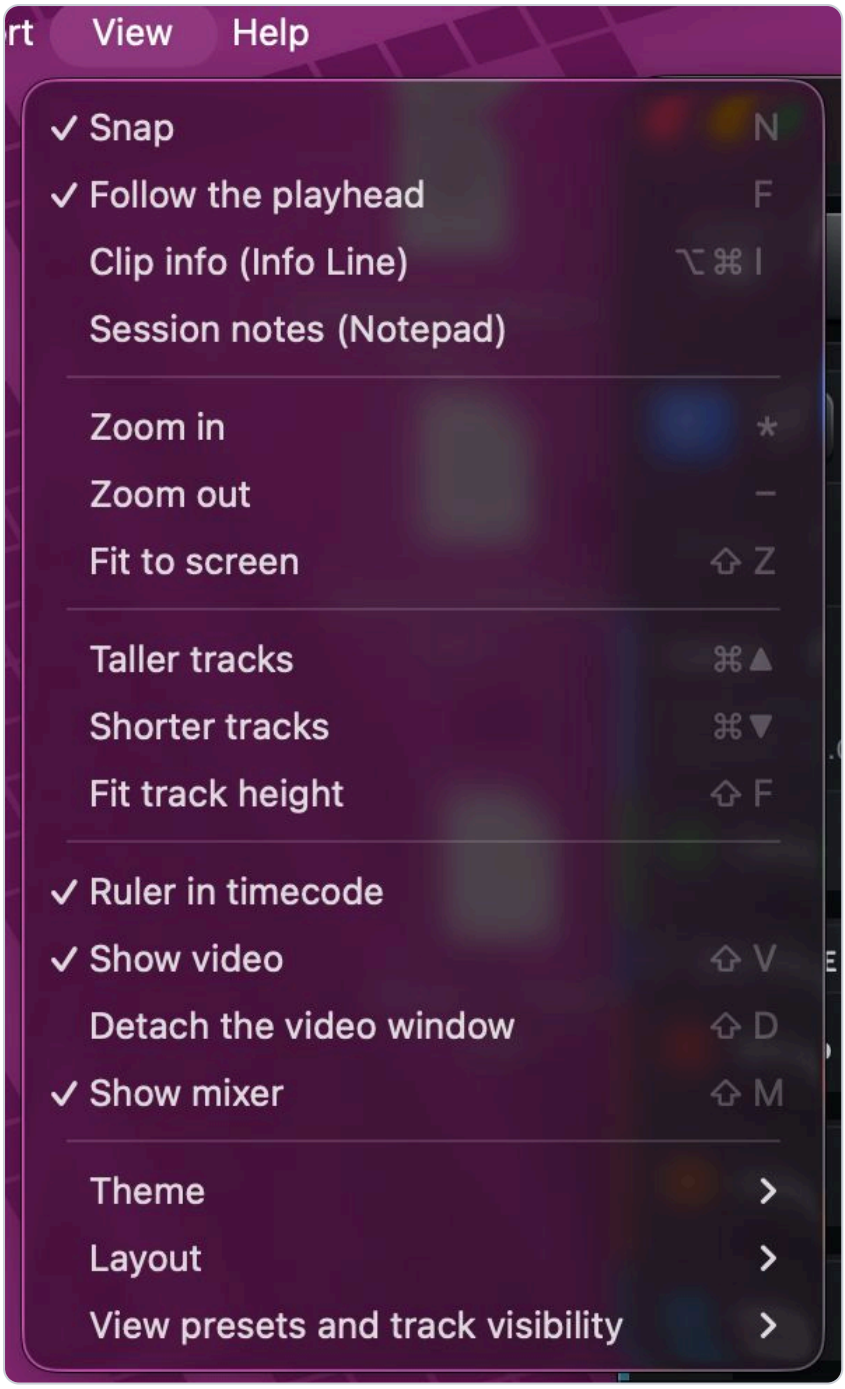
The ADR menu

Command	Shortcut
ADR window	`Cmd/Ctrl+Shift+A`
Stops the recording of the cue	`Cmd/Ctrl+Shift+R`
Next cue	`Cmd/Ctrl+Shift+→`
Previous cue	`Cmd/Ctrl+Shift+←`



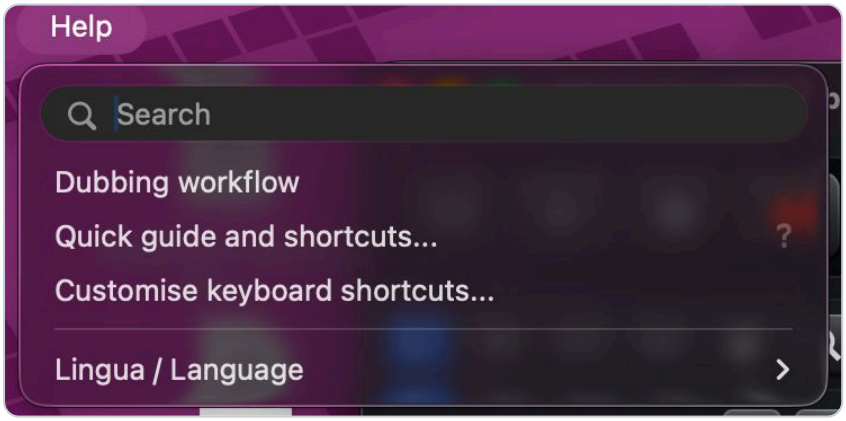
The Transport menu

Command	Shortcut
Stop	`Space`
Go to start	`Home`
Go to end	`End`
Loop	`L`
Loop from selection	`Shift+L`
Back one frame	`←`
Forward one frame	`→`
Back one second	`Shift+←`
Forward one second	`Shift+→`
Stop recording	`R`
Monitor the input while recording	`I`



The View menu

Command	Shortcut
Snap	`N`
Follow the playhead	`F`
Zoom in	`=` / `+`
Zoom out	`-`
Fit to screen	`Shift+Z`
Show mixer	`Shift+M`
Show video	`Shift+V`
Clip info (Info Line)	`Cmd/Ctrl+Alt+I`
Dock the video window	`Shift+D`
Fit track height	`Shift+F`
Advance the video by 1 ms	`,`
Delay the video by 1 ms	`.`
Taller tracks	`Cmd/Ctrl+↑`
Shorter tracks	`Cmd/Ctrl+↓`



The Help menu

Command	Shortcut
Quick guide and shortcuts...	`?`

4. Sessions and files

4.1 Creating, opening and saving

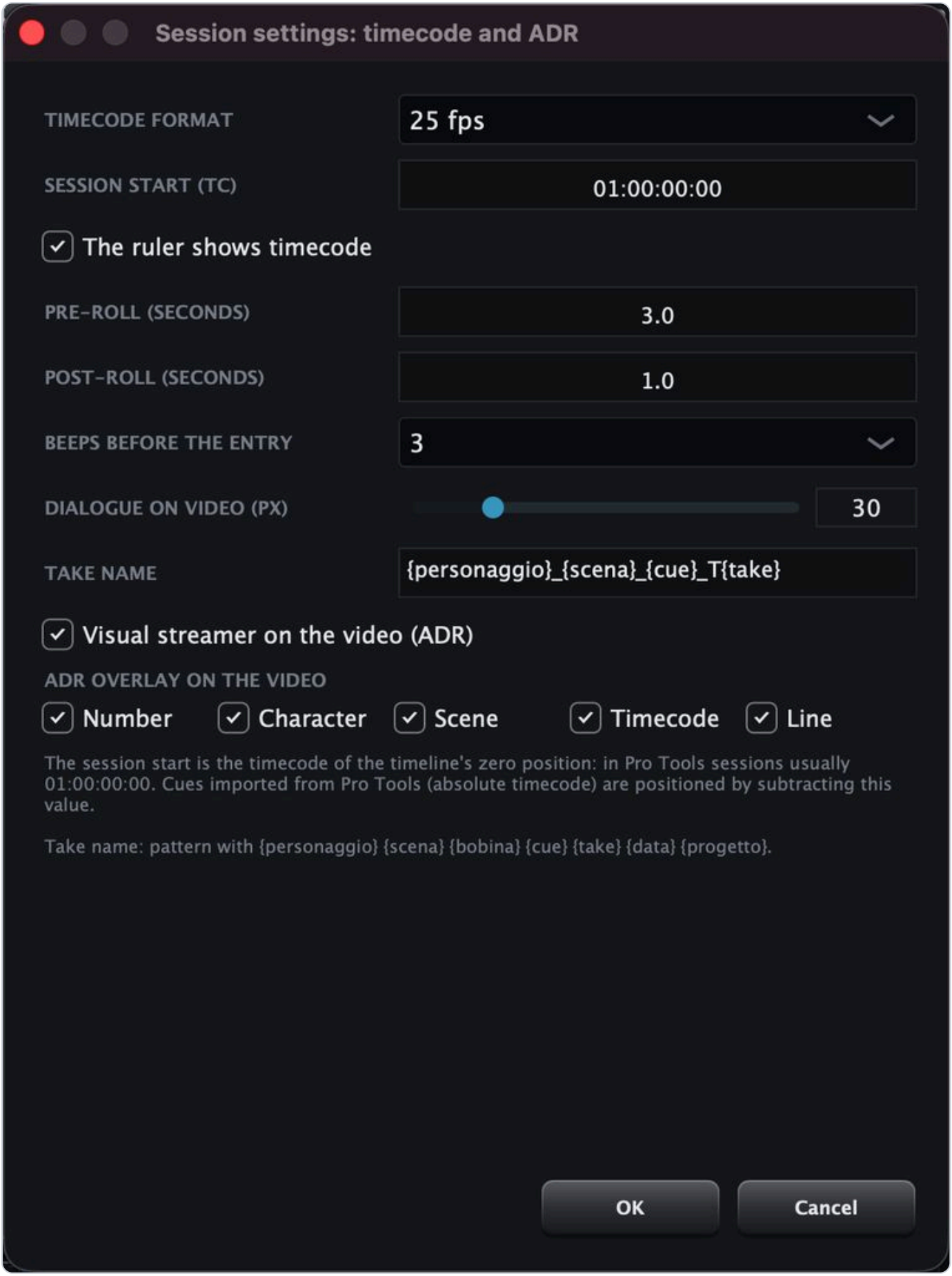
A **session** is a file with the `.adb` extension holding tracks, clips, cues, markers, mixer settings and the paths of the audio and video files. It **does not contain the audio**: the audio files stay where they are (or in the session's media folder). `.arostudio` sessions from earlier versions can still be opened.

- **File > New session** (Cmd + N): an empty session, with no tracks. **New from template** creates the right tracks at once: *Standard dubbing* (Guide, M&E, Effects and a Dialogue track, each on its own bus), *ADR recording only* (Guide and ADR track, M&E as a headphone reference) and *Review and mix* (Dialogue, M&E and Effects for the final mix).
- **Open** (Cmd + O), **Open recent**, **Save** (Cmd + S), **Save as** (Shift + Cmd + S). Opening a file from the Finder or dragging it onto the icon also works with several audio and video files together.
- An asterisk in the window title means unsaved changes; on closing, the program asks what to do.

4.2 Where recordings and processed files go

- **Recordings** (the ADR takes and recorded tracks) are written to the `<session name>_media` folder next to the session file, so the project moves and archives together with its files. Until the session is saved they go to `Music/ADR Dub Studio/Registrazioni`; on the first *Save as* the program offers to copy them next to the session.
- Files created by processing (time-stretch, Time Warp, JOIN, Sample Editor, Freeze, offline processing) go to `Music/ADR Dub Studio/Elaborati`. The original is never modified: every processing creates a new file and can be undone with Undo.
- **Collect the media in the project folder** (File) copies into one folder all the files used by the session, to move it to another computer.

4.3 Session settings

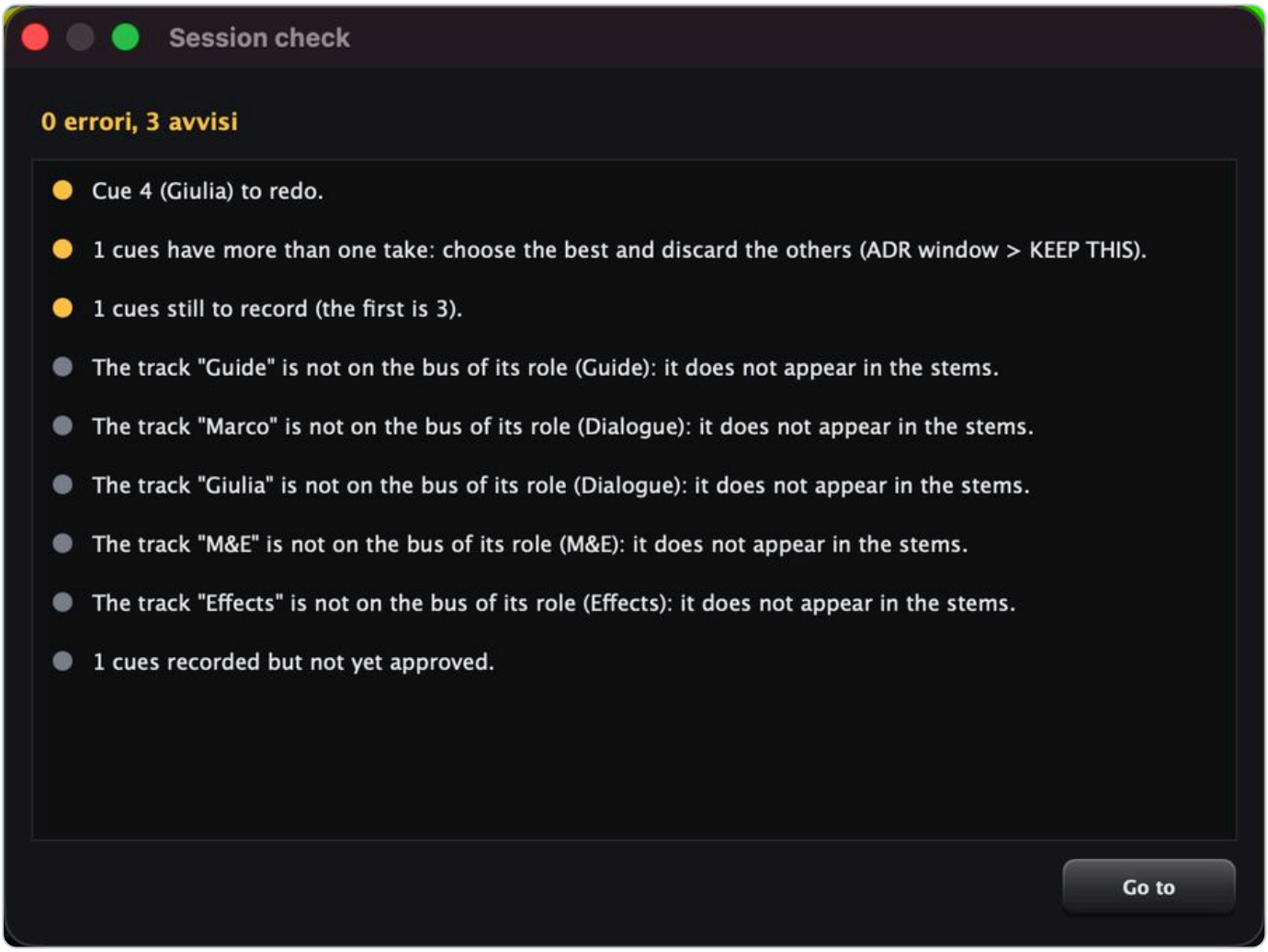


File > Session settings: timecode and ADR.

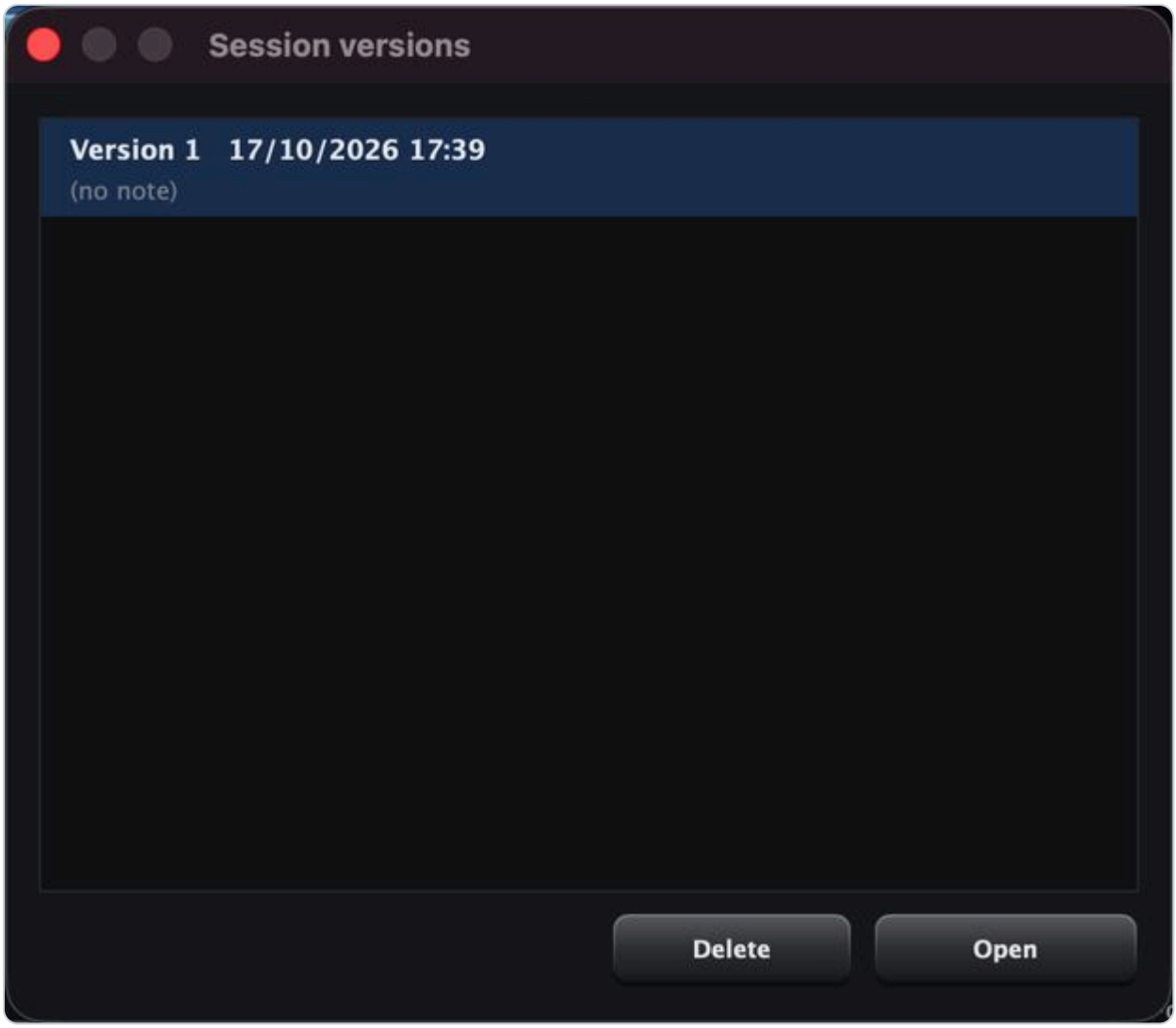
Setting	What it does
Timecode format	23.976, 24, 25, 29.97 (non-drop and drop frame), 30, 50, 60 fps. When you import a video the format takes the video's fps by itself.
Session start (TC)	The timecode of the timeline's zero position: in Pro Tools sessions usually 01:00:00:00. Cues imported with absolute timecode are positioned by subtracting this value.
The ruler shows timecode	Ruler in timecode or in minutes and seconds.
Pre-roll / Post-roll	Seconds of playback before the cue's in point and after its out point when recording.
Beeps before the entry	How many countdown beeps (0–5), one second apart, before the cue's in point.
Dialogue on video (px)	Size of the line of dialogue in the video overlay.
Take name	Pattern for the names of recorded files with placeholders `{personaggio}` (character), `{scena}` (scene), `{bobina}` (reel), `{cue}`, `{take}`, `{data}` (date), `{progetto}` (project). The placeholders keep their Italian names also in the English interface.
Visual streamer on the video	The bar that slides over the picture during the countdown and the cue.
ADR overlay	Which data to write on the picture: number, character, scene, timecode, line.

4.4 Versions, checking and recovery

- **Save a version of the session** (File): writes a numbered copy with date and a note into a folder next to the session (`<name>_versioni`). **Session versions** lists the versions: *Open* loads one, *Delete* removes it. Use it before risky operations such as a ReConform.
- **Check the session before delivery** (File): lists errors and warnings (missing media, clips on tracks without a role, cues still to record or to redo, cues marked recorded but without takes, several active takes, video whose fps differ from the timecode…). *Go to* takes you to the problem.
- **Session notes** (View): a notepad saved with the session.
- **Relink missing media** (File): if a folder is moved, clips stay as *offline* clips (hatched, labelled OFFLINE) and are not lost. By pointing to a folder, the program searches by file name (also in subfolders) and relinks everything in one undoable step.



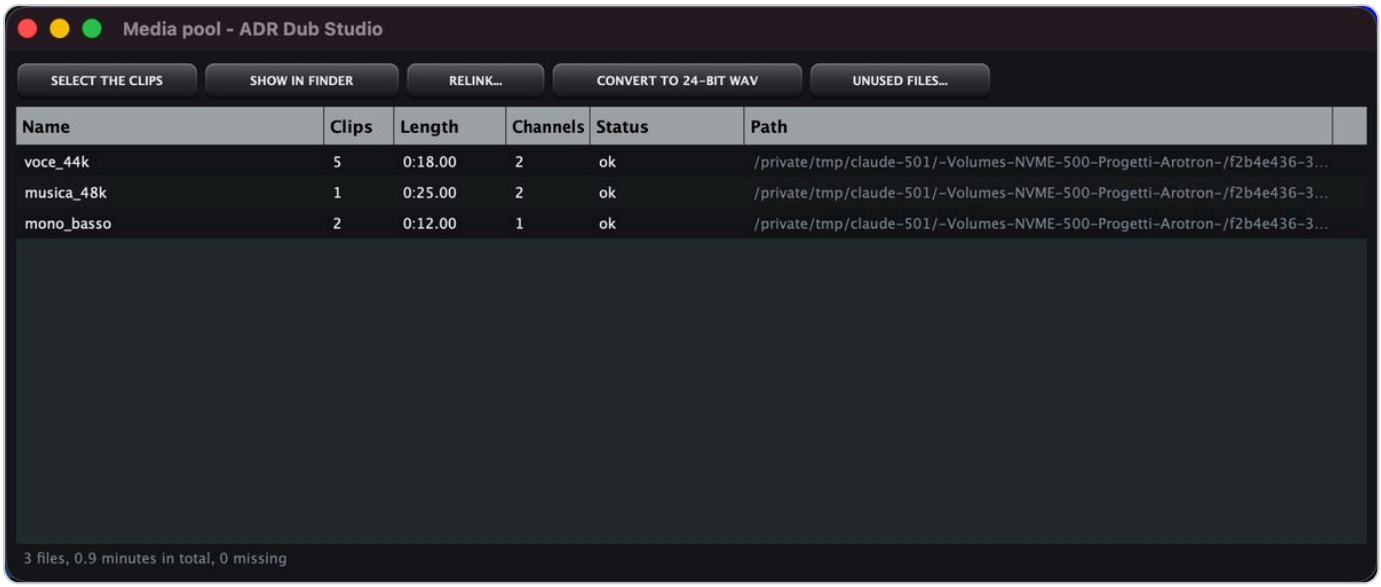
The session check: yellow warnings and grey information.



The session versions.

4.5 The media pool

File > Media pool shows every file used by the session, with the number of clips that use it, the length, the channels and the status (*ok*, *in memory* if generated by the program, *MISSING*).



The media pool.

- **SELECT THE CLIPS**: selects in the timeline all the clips that use the chosen files.
- **SHOW IN FINDER**: shows the file in Finder (Explorer on Windows).
- **RELINK…**: same as File > Relink missing media.
- **CONVERT TO 24-BIT WAV**: writes a 24-bit WAV copy at the session rate in the media folder and uses it in place of the chosen file (undoable).
- **UNUSED FILES…**: lists the files in the media folder that no clip uses (for example discarded takes) and can move them to the trash.

5. Importing audio and video

- **Audio**: *File > Import audio* (Cmd + I) or drag the files onto the timeline. Each file goes onto a new track, at the session start. Files are read into memory and converted to the session rate; the formats read are WAV, AIFF, FLAC and those supported by the system.
- **Video**: *File > Import video* (Shift + Cmd + I) or drag a video onto the timeline. The video appears in its lane and in the viewer; its audio is extracted and put on a **Guide** track. The video's fps become the session timecode's.
- **Cues**: dragging a cue file (CSV, TSV, EDL, SRT, WebVTT, or a Pro Tools marker list) onto the tracks starts the ADR import (see *The ADR workflow*).
- **Takes already recorded**: *ADR > Import recorded takes from a folder* matches them to the cues by number (see *The ADR workflow*).

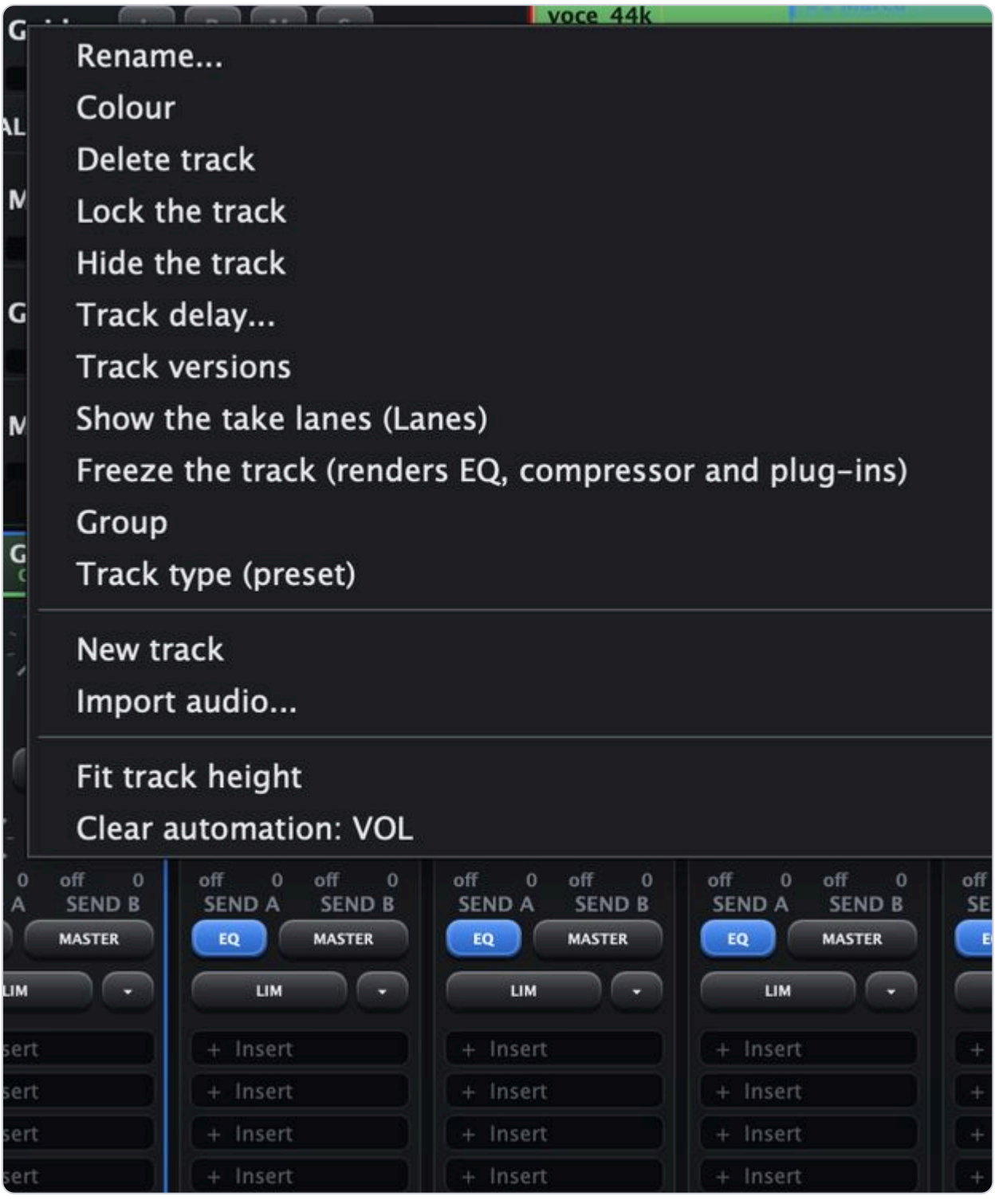
> **Tip** For a long video, extracting the audio and thumbnails can take a moment: a progress bar at the bottom shows it. Sessions open faster after the first time because thumbnails are kept.

> **Note** **Video offset**: if the picture lags behind the audio, adjust it with the –F/+F and –/+ buttons under the viewer, by dragging the video lane, or by typing the milliseconds.

6. Tracks

6.1 The track menu

Right-clicking a track header opens its menu. If several tracks are selected (Cmd + click or Shift + click on the headers) the menu acts on all of them.



A track's context menu.

Item	What it does
Rename… / Colour / Delete	Name, colour (eight colours) and deletion of the track.
Lock the track (Lock)	The track's clips can no longer be moved or edited (useful to protect finished work). The header's **L** button does the same.
Hide the track	The track disappears from the timeline and the mixer but keeps playing. The **V** button at the top left shows it again.
Track delay…	Delays the whole track by 0–500 ms (to align a distant microphone or compensate an external plug-in); it also applies to the export.
Track versions	Alternative versions of the track's clips (see below).
Show the take lanes	Lays out overlapping clips on separate lanes, with the active lane highlighted; it is for choosing between several takes.
Freeze the track	Renders the track with EQ, compressor and plug-ins into an audio file: from then on the file plays and uses no CPU. It can be unfrozen.
Group	Puts the track into a track group, creates a new one, or creates the "Dialogue" group with all the Dialogue tracks and the character tracks.
Track type (preset)	Applies the role and the recommended chain (see below).
New track / Import audio / Fit height / Clear automation	Quick commands.

6.2 Track roles and buses

The **track type** assigns a role and loads a starting chain (4-band EQ, compressor, limiter, level) designed for that role; the values are a starting point to be adjusted by ear on the material. Each role has its own group **bus**, used to export the stems.

Role	Chain
Dialogue	High-pass 80 Hz, –2.5 dB at 250 Hz (body and boominess), +2.5 dB at 3.5 kHz (intelligibility), +1.5 dB of air above 10 kHz. Compressor 3:1, threshold –20 dB, attack 10 ms, release 120 ms. Limiter at –3 dB.
M&E	Music and effects without dialogue. High-pass 30 Hz, –1.5 dB at 300 Hz, –2 dB at 2.8 kHz (makes room for the dialogue), +1 dB of air. Gentle compressor 2:1, threshold –18 dB, attack 30 ms, release 200 ms. Limiter at –1 dB. Level –6 dB.
Effects	High-pass 40 Hz, –1.5 dB at 350 Hz, +1.5 dB at 5 kHz (presence), +1 dB of air. Compressor 4:1, threshold –14 dB, attack 5 ms, release 80 ms. Limiter at –2 dB. Level –3 dB.
Guide	Original reference dialogue: no processing, level –6 dB, Guide bus. It is excluded from deliveries and from the Dialogue/M&E/Effects buses; automatic alignment uses this track as the model.
None	Everything off: EQ, compressor and limiter, level at 0 dB.

6.3 Track groups

Track groups (for example "Dialogue" with the character tracks) have a header with a triangle: one click collapses it and hides the tracks, double-click renames it, the **M** and **S** buttons act on all its tracks. To create one: select several tracks (Cmd + click on the headers) and choose *Group > New group with the selected tracks*. Do not confuse them with the mixer's **group buses** (a track's output).

6.4 Versions, lanes and take comping

- **Track versions**: each track can hold several versions of its clips (for example "Edit A" and "Edit B"). The *Track versions* submenu creates an empty version, duplicates the current one, renames, deletes; *Lanes from versions* shows them all as lanes and *Versions from lanes* does the opposite.
- **Lanes**: with *Show the take lanes* overlapping clips go on different lanes. Right-clicking a clip offers *Solo Lane*: listen only to that clip's lane.
- **Comp tool** (C): on a track with several overlapping takes, drag a range on a clip: that take becomes the one playing there and the others are muted. In this way you can assemble the best parts of several takes.

6.5 Reference track and A/B comparison

Import the comparison mix (for example the film's original version) onto a track and choose **Track > Reference track**: the track shows "[REF]" in its name, **does not play and is not exported**. With **Track > A/B comparison with the reference** (the `` ` `` key) listening switches to the reference tracks only and an amber frame with the text "A/B: REFERENCE" reminds you where you are; press again to return to the mix. The reference track's fader is used to match levels.

6.6 Mixer snapshots and track presets

- **Mixer snapshots** (Track menu): saves the state of faders, pan, mute, solo, sends, input gain, delay and output of every track, plus master, returns and buses. *Save the current mixer state…* creates one; choosing a name recalls it; *Undo the last recall* goes back; *Delete a snapshot*. Snapshots are saved with the session and attach to the tracks by position.
- **Track presets** (Track menu): *Save the selected track as a preset…* records level, pan, input gain, EQ, compressor, limiter, sends, delay, type and output (not the plug-ins); presets are in `~/Library/ArotronStudio/TrackPresets` and are applied to the selected track(s) by choosing their name.
- **Track archives** (Track menu): *Export the selected tracks as an archive* creates a package with tracks, clips and audio files to take into another session; *Import a track archive* adds it to the current session.

6.7 View presets and track visibility

View > View presets and track visibility saves the **zoom and track height** with a name (for example "Detail" and "Overview") and the **visibility configurations** (which tracks are visible, for example "Dialogue only"); a click on the name recalls them. They are saved with the session.

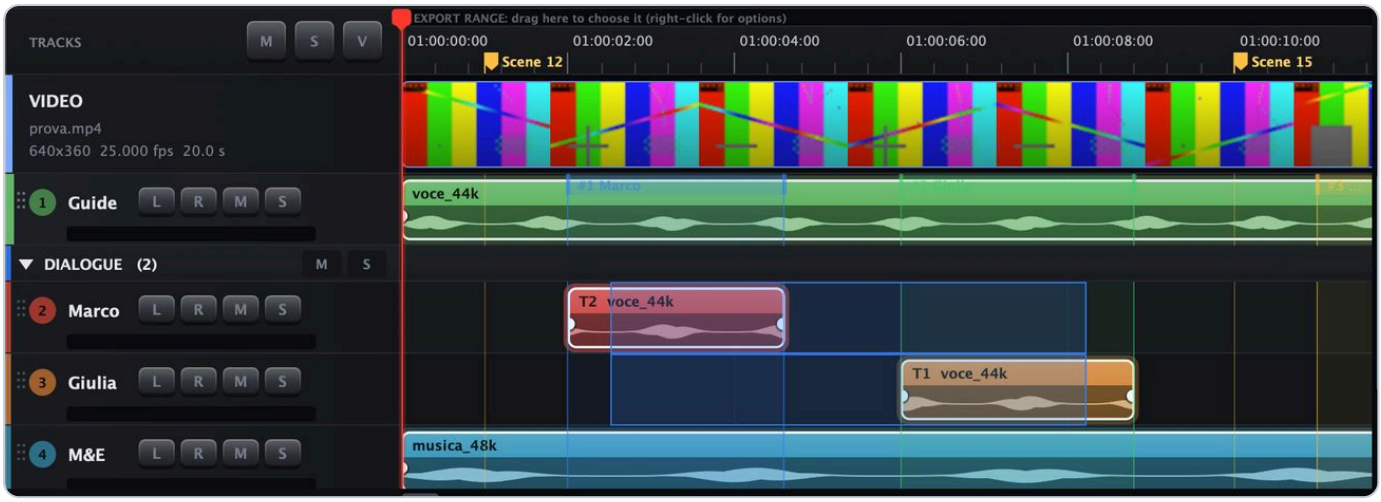
7. Clips and editing

7.1 Selecting, moving and trimming

- **Moving**: drag a clip with the Select tool, also onto another track. With **SNAP** on, edges snap to other clips' edges, the playhead, the loop region, cue in/out points, markers and the export range limits. The clip's sync point (if any) snaps like an edge.
- **Trimming**: drag a clip's left and right edges. The original file is not changed; the clip just shows a portion.
- **Fades**: the two white round handles at the top of a selected clip set the fade-in and fade-out.
- **Nudge**: the arrow keys move the playhead by one frame (with Shift by one second). Q and W trim the start and the end of clips at the playhead.
- **Split** (S) splits the selected clips at the playhead; **Duplicate**, **Delete**, **Ripple delete** (closes the gap).

7.2 The tools one by one

Tool	How to use it
Select (V)	Selects, moves and trims clips. **Cmd + drag** on a track: selects the portion and loops it. On cue edges (the strip at the top of the lanes): drag an edge to change in or out, the body to move the cue.
Razor (B)	Click a clip to split it (with snapping). **Shift**: splits all tracks.
Slip (Y)	Drag to slide the content inside the clip without moving it.
Stretch (E)	Drag a clip edge to lengthen or shorten it: the audio is recalculated (WSOLA) **without changing pitch**, between 25 % and 400 %. The edge snaps to cues; the waveform shows a preview while dragging.
Hand (H)	Drag to move the view.
Zoom (Z)	Click to zoom in, Alt + click to zoom out, drag to zoom on an area.
Range (X)	Drag a time range over one or more tracks; then, from the Edit menu: split at the edges, delete (leave the gap), remove (close the gap), keep only the range, loop it.
Mute (K)	Click a clip to mute or unmute it (on an ADR take it makes it the active take).
Eraser (G)	Click a clip to delete it. In Ripple or Shuffle mode the gap closes.
Comp (C)	See *Versions, lanes and take comping*.
Time Warp (O)	See *Time Warp*.
Clip envelope (D)	See *Clip gain envelope*.
Scrub (J)	Drag over the tracks to hear the position in short bursts, like moving the tape by hand; it also works with Alt + drag on the ruler.
Automation (U)	Click a track to add a point to the displayed curve (VOL, PAN, SEND A or SEND B: change it with the "AUTO/VOL" button in the header), drag to move it, right-click to delete it, drag the line between two points to move it vertically.



The Range tool: the chosen range over several tracks.

7.3 Edit modes

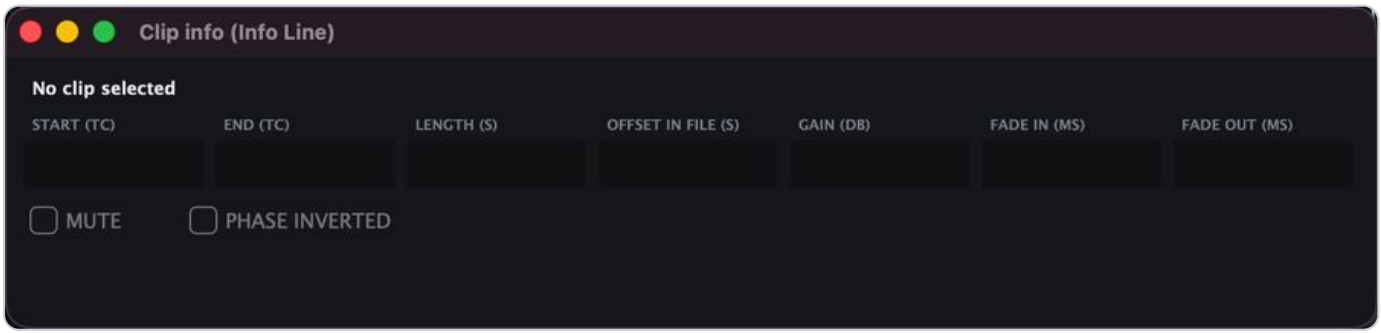
The **NORMAL** button (on the right, in the edit button bar) cycles on each click through three modes: **Normal** (deleting leaves the gap), **Ripple** (deleting closes the gap on the track) and **Shuffle** (moves too: neighbouring clips slide to make room).

7.4 Crossfades, snapping and automatic fades

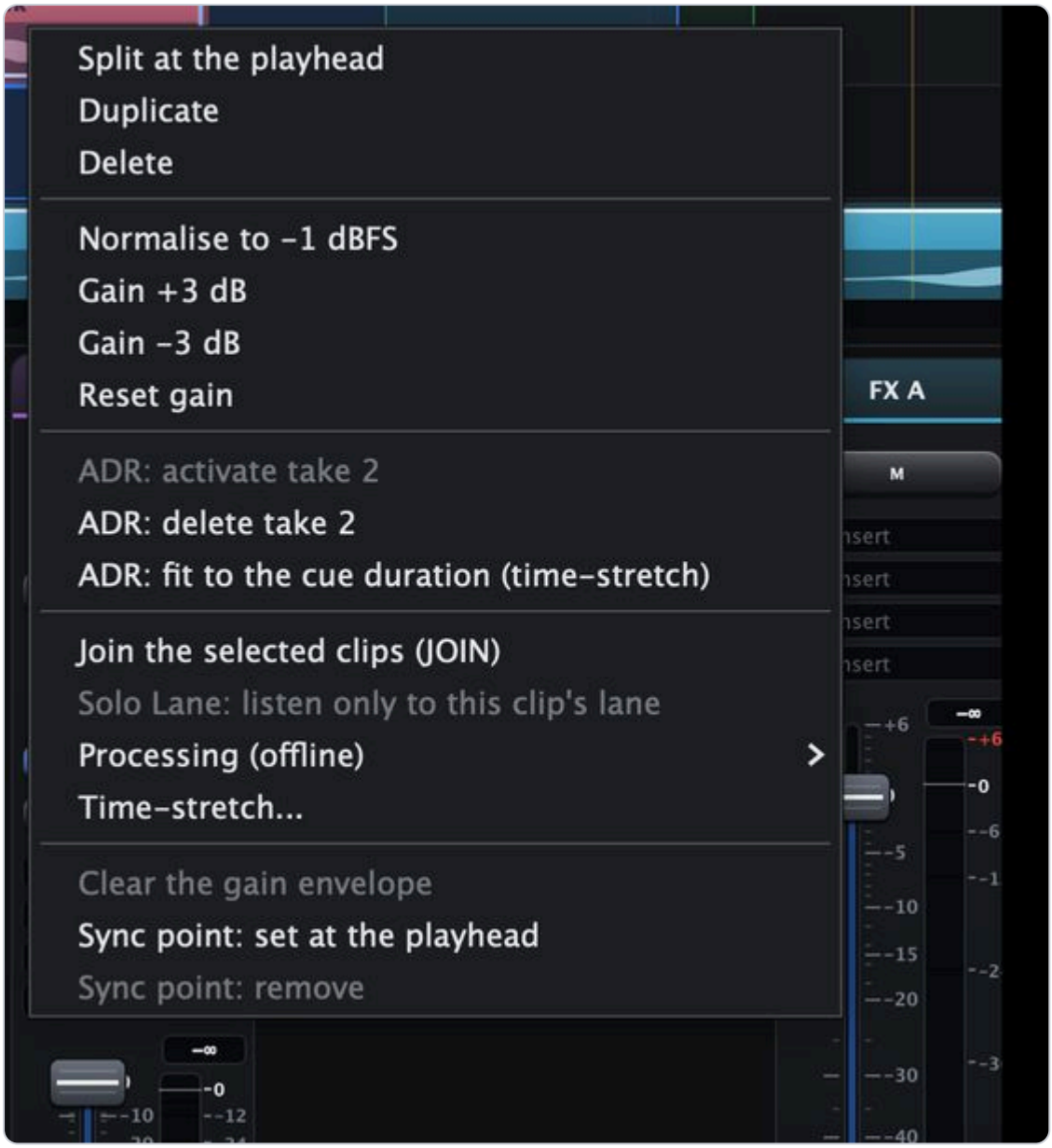
- **Automatic crossfades**: when two clips overlap on the same track, the overlap becomes a crossfade (constant power, or constant gain if it is the same continuous material).
- **Edit > Crossfade between two selected clips…**: creates an overlap of a chosen length across the cut, with automatic, linear or constant-power shape.
- **Cuts on the zero crossing** (Snap to Zero Crossing): cuts move to the nearest zero crossing, to avoid clicks.
- **Automatic fades**: a short fade at the edges of new clips and cuts (the command changes the length at every click, or turns it off).

7.5 Gain and Info Line

The right-click menu of a clip offers **Normalise to –1 dBFS**, **Gain +3 dB**, **–3 dB** and **Reset gain**. To type exact values open **View > Clip info (Info Line)**: it shows and edits start and end (timecode), length, offset in the file, gain, fade in/out, mute and inverted phase of the selected clip.



The Info Line window.



A clip's context menu.

7.6 JOIN, silences and offline processing

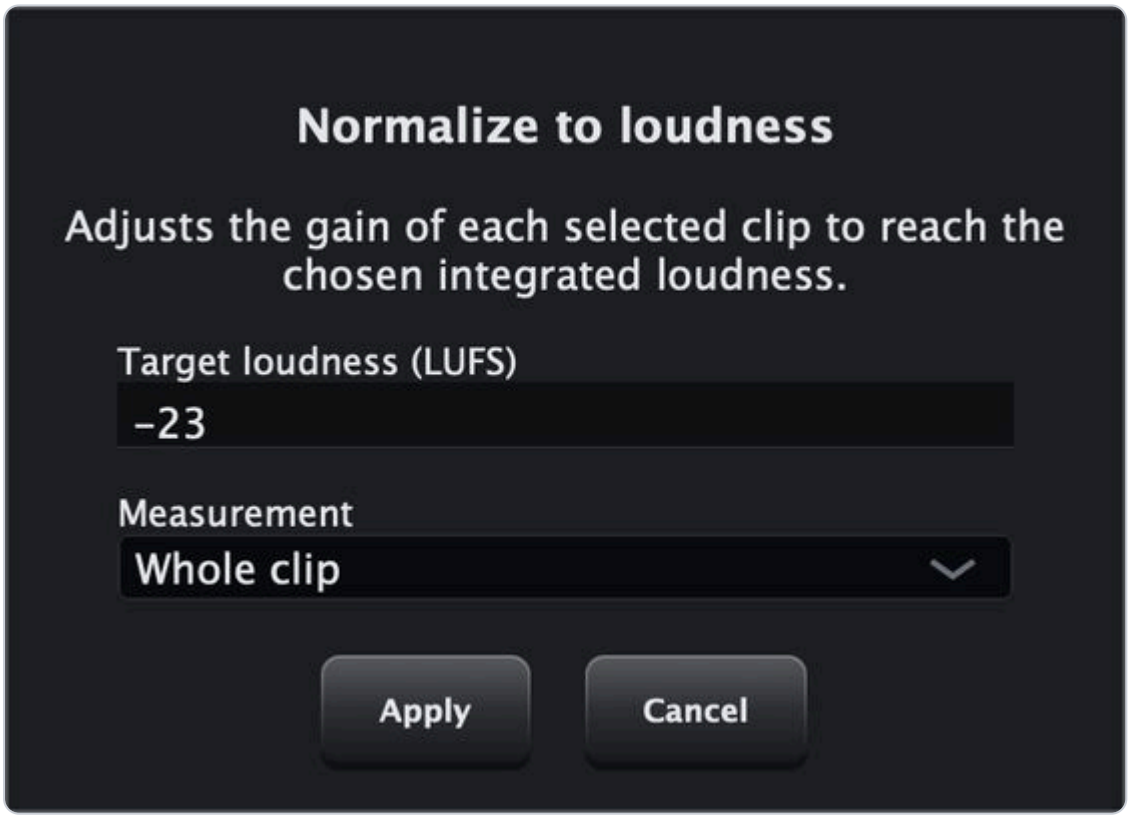
- **Join the selected clips (JOIN)** (Cmd + J): the selected clips of a track (at least two, not muted) become one clip in a new file, with gaps as silence and with fades, crossfades and gain already applied. One undo step.
- **Detect / strip silence** (Edit): within the chosen clips it finds the pauses (threshold, minimum pause, margin) and splits at the edges or removes the silences (Strip Silence).
- **Offline processing** (Direct Offline Processing, Edit menu or right-click a clip > Processing): *invert the phase*, *reverse*, *remove the DC offset* and *normalise to loudness* (to a LUFS target, measuring the whole clip or only the speech).
- **Time-stretch** (Cmd + Alt + T or right-click): changes the length of the selected clips as a percentage (25–400 %) keeping the pitch (voice) or together with the pitch (varispeed); for an ADR take it can be fitted to the cue's exact length.



Detect / strip silence.



Time-stretch.

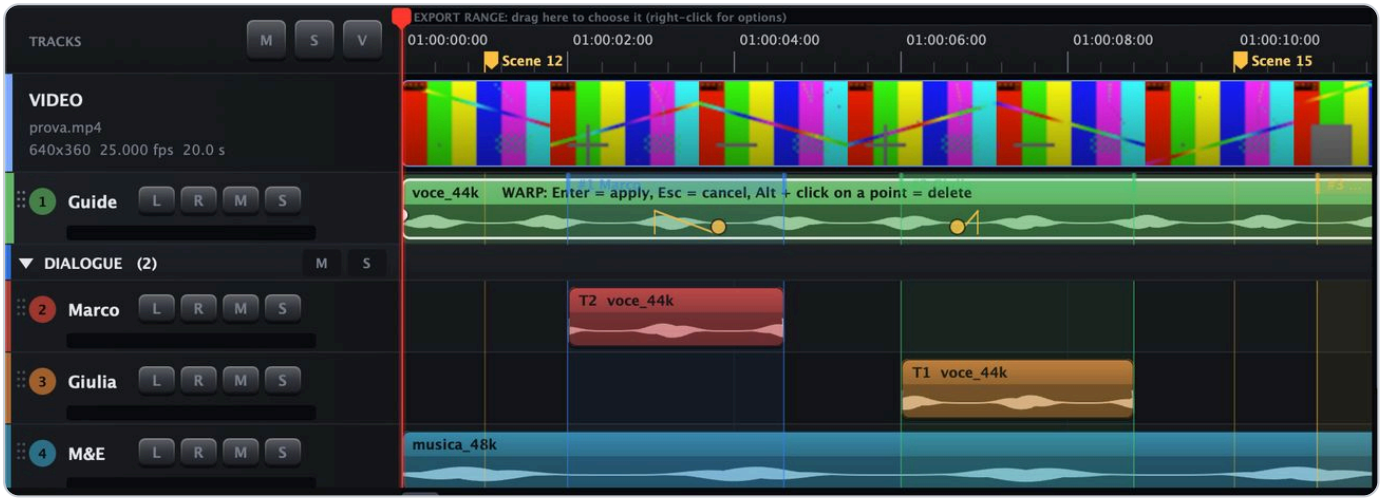


Normalise to loudness.

7.7 Time Warp: moving words inside a take

The **Time Warp** (tool **O**) lets a word land exactly where you want without changing the speed of the whole take. It works with anchor points: the material between two points is stretched or squeezed (pitch unchanged) and the material after the last point follows the shift.

1. Choose the Time Warp tool (key **O**).
2. Click a clip at the word to move: an anchor point (yellow circle) appears, with the label at the top of the clip.
3. Drag the point to where the word should land (snapping to cues and the playhead). You can add more points: those not moved serve as fixed anchors.
4. **Return** applies, **Esc** cancels; **Alt + click** on a point removes it. The result is a new file in the Elaborati folder and one undo step.



The Time Warp with two anchor points before applying.

> **Note** Points must be at least 60 ms apart and the ratio between two points stays between 1/4 and 4. Anything outside is corrected automatically.

7.8 Sync point

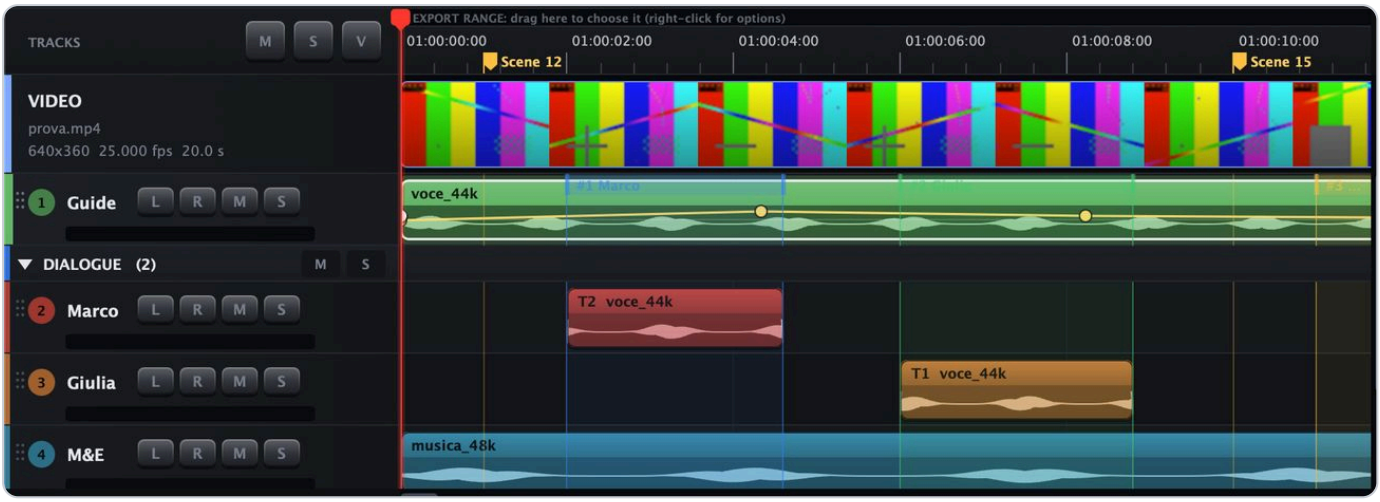
A **sync point** marks inside a clip the sound that should coincide with something (for example a strong consonant of a take with the lip movement, or the cue's in point). Put the playhead on the sound, choose **Edit > Sync point: set at the playhead** (Alt + S): a light-blue flag appears on the clip. Then:

- when dragging the clip the sync point snaps to cues, markers and the playhead as if it were an edge;
- **Sync point: move onto the cue (or the playhead)** (Shift + Alt + S) moves the clip so that the point falls on the entry of its ADR cue; for a clip with no cue, on the playhead position;
- **Sync point: remove** deletes it. The point is kept when trimming, splitting or sliding the clip and follows the material after a Time Warp or a stretch.

7.9 Clip gain envelope

With the **Clip envelope** tool (key **D**) you draw a gain curve inside the clip: it is for lowering a breath or a gust of wind just at that spot, without automating the whole track. At the top of the clip it is +12 dB, the light line is 0

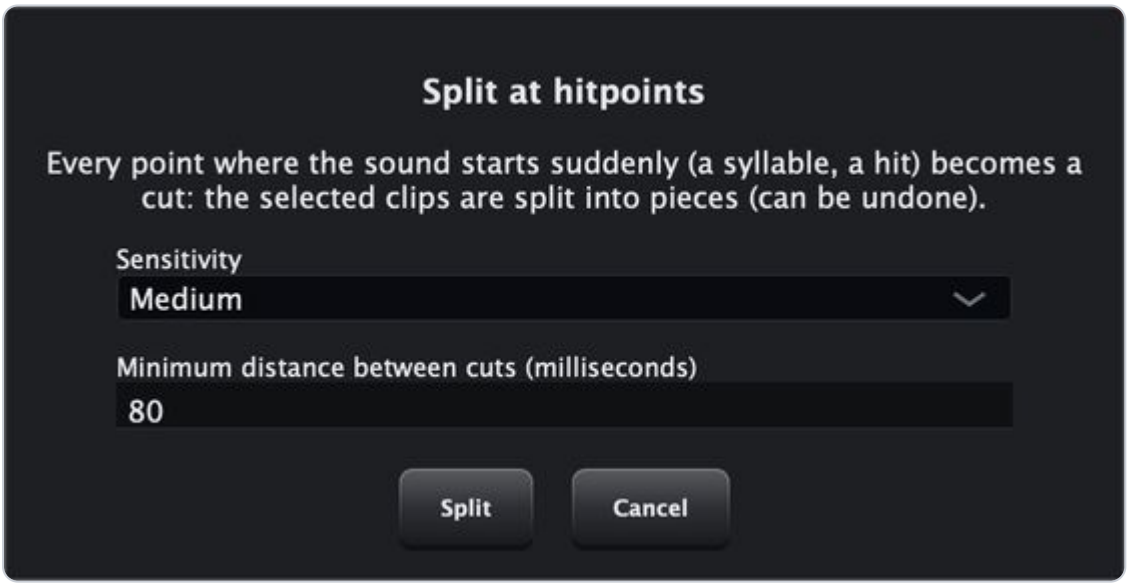
dB, at the bottom silence. Click to add a point, drag to move it, **Alt + click** removes it; *Clear the gain envelope* (right-click or Edit menu) resets it. The envelope follows the content when you trim or slide the clip, and applies in playback, export and JOIN.



The clip envelope with two points.

7.10 Hitpoints and slicing

Edit > Split at hitpoints… finds where the sound starts suddenly (a syllable, a hit) and splits the selected clips into one piece per attack, in one undo step. You choose the **sensitivity** (low, medium, high) and the minimum distance between cuts. In the Sample Editor the same points can be seen before splitting.



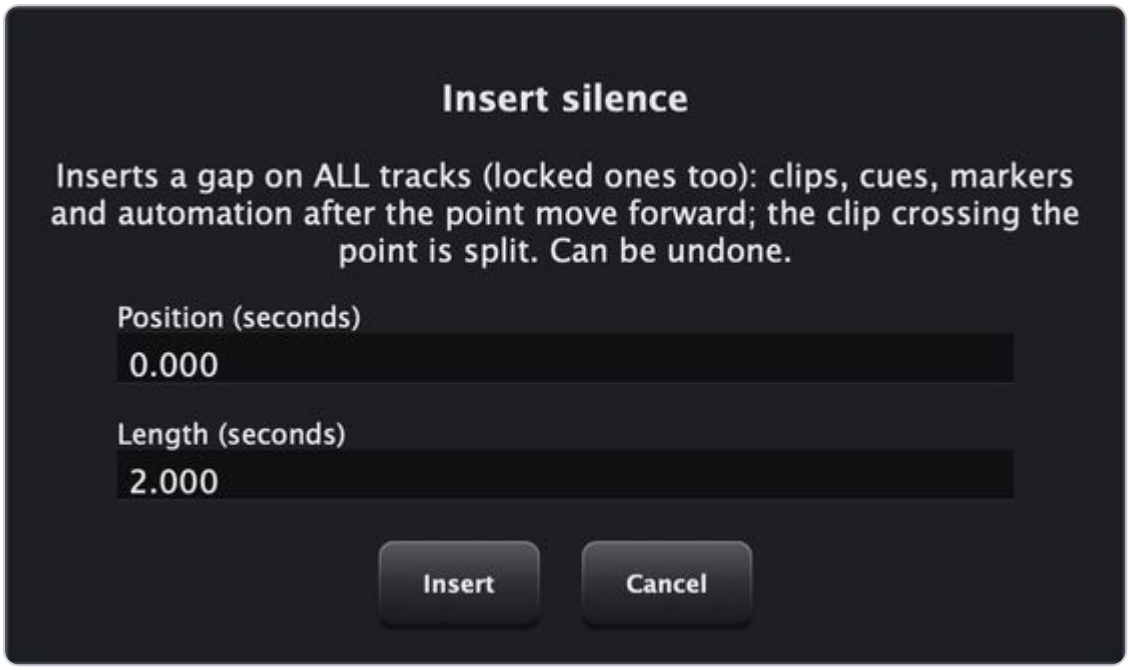
Split at hitpoints.

7.11 Insert silence and remove time

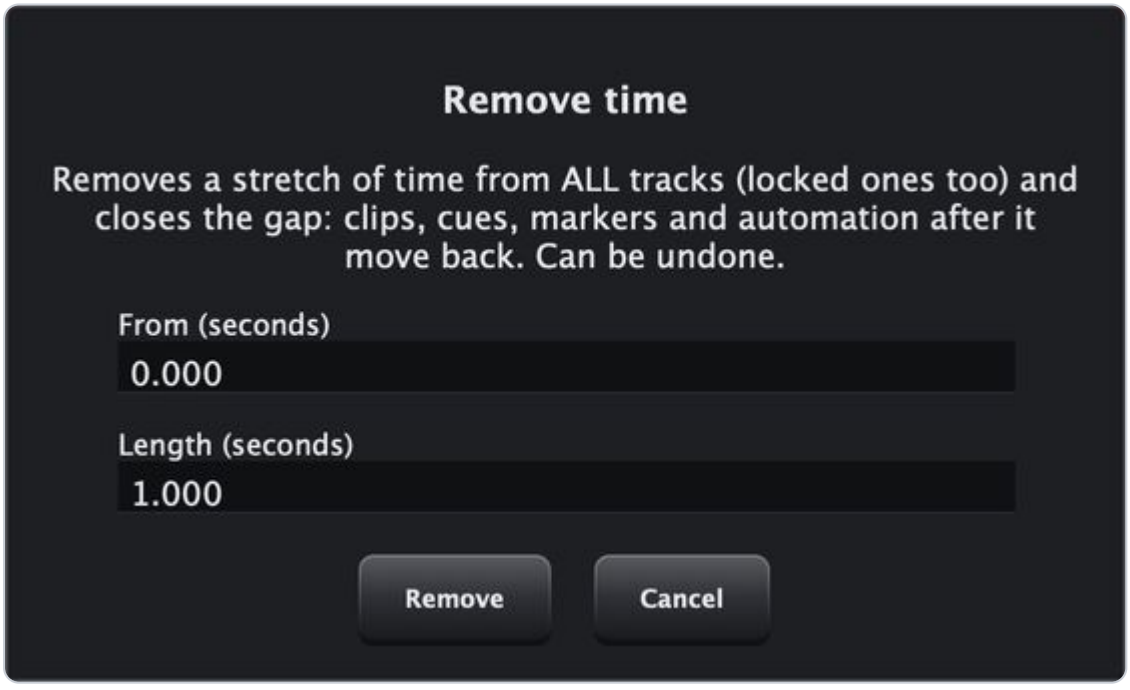
- **Insert silence** (Edit): inserts a gap of a chosen length at a position; on **all tracks** (locked ones too) the clips after the point slide forward and the clip crossing the point is split. Cues, markers, automation and the export range move too.

- **Remove time** (Edit): removes a stretch of time from all tracks and closes the gap; by default it proposes the range already chosen (Range, loop region or export range).

> **Note** If there are frozen tracks the program asks you to unfreeze them first: frozen audio cannot be shifted. One undo reverts the operation.



Insert silence.

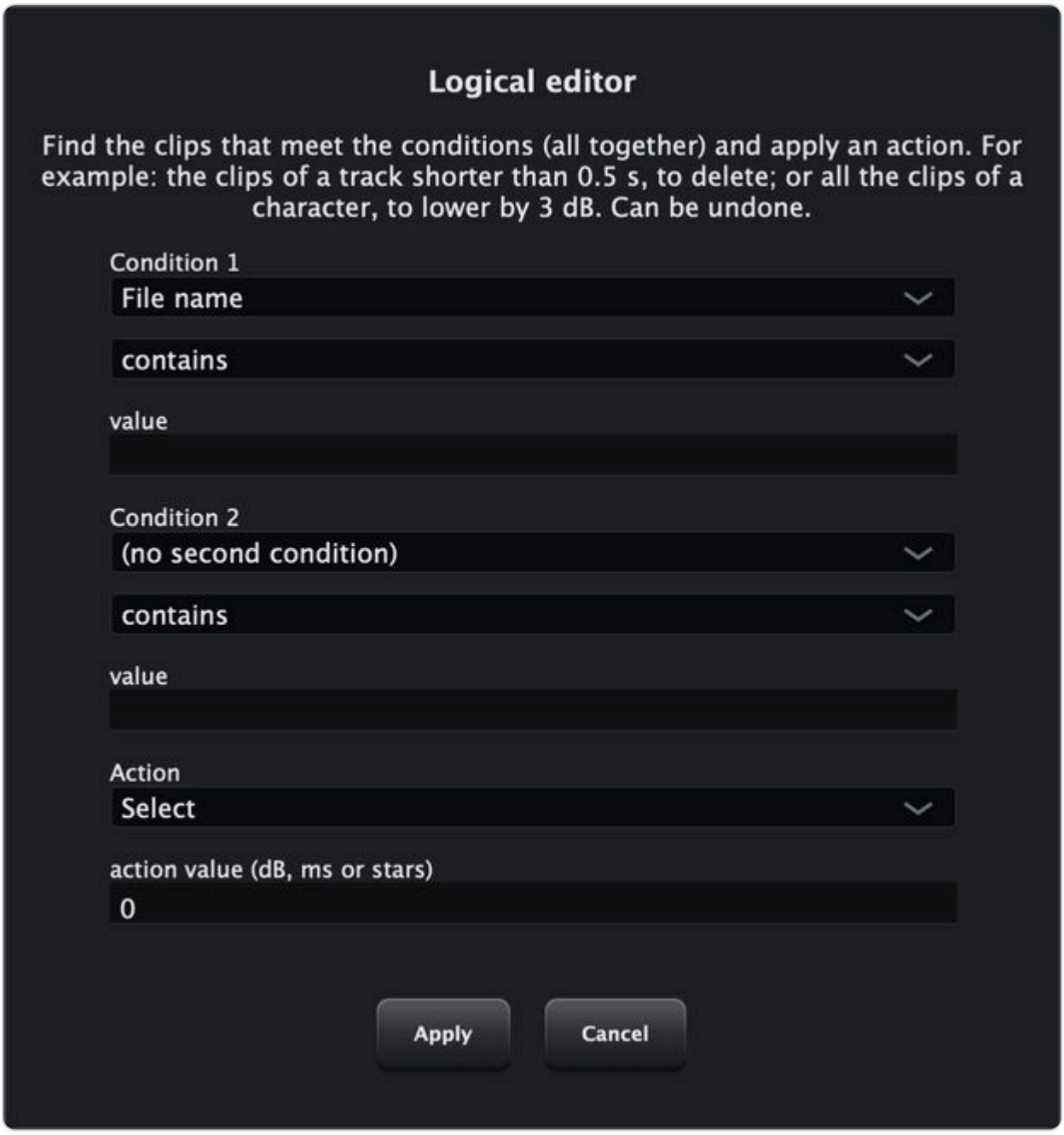


Remove time.

7.12 Logical editor and batch rename

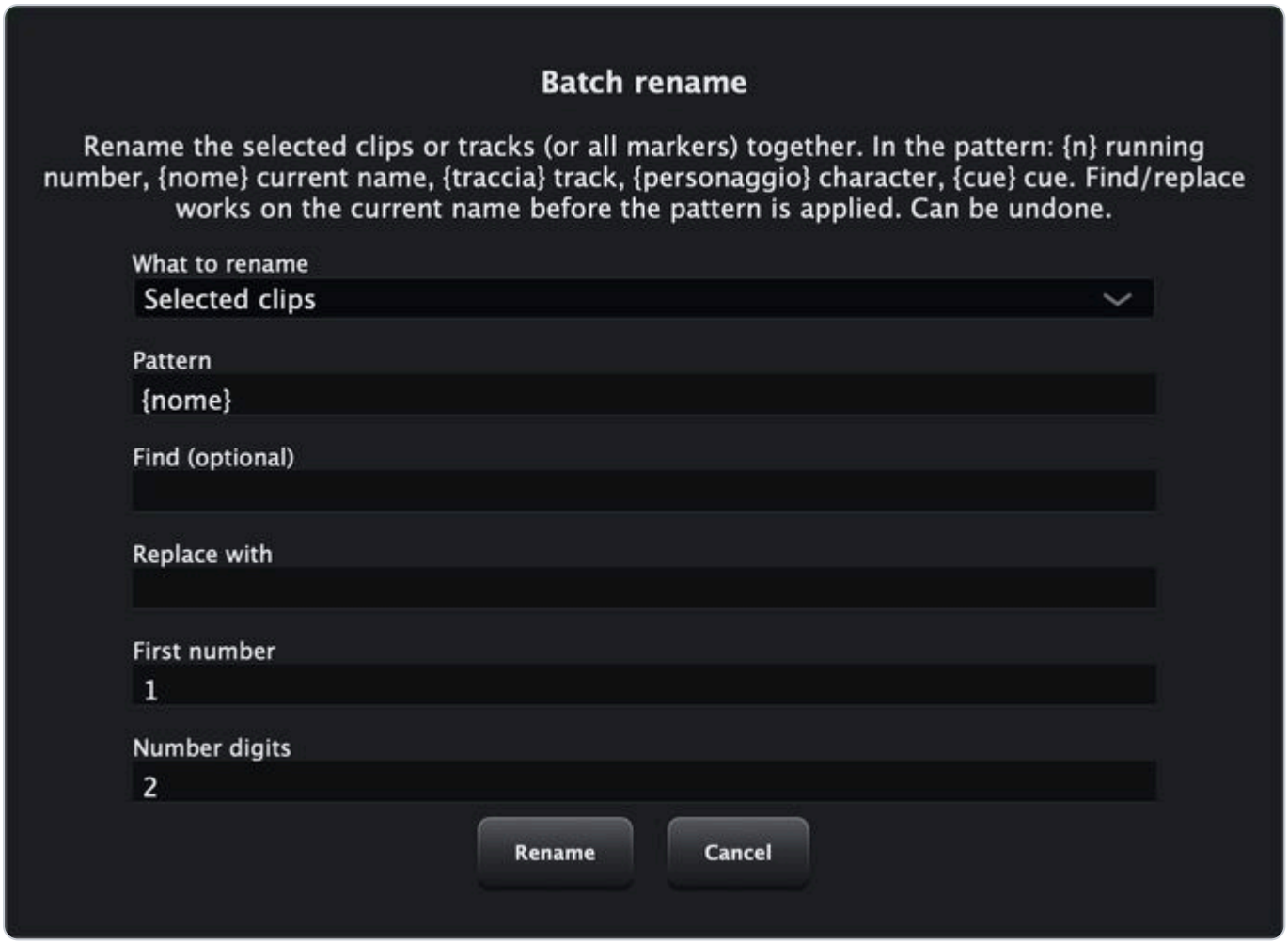
Edit > Logical editor… selects or modifies clips according to one or two conditions (file name, track name, cue character, length, gain, muted, take stars, position) with the operators *contains*, *equals*, *is greater than*, *is*

less than. The actions are: select, mute, unmute, delete, set or add to the gain, set the fades, rate with N stars, reject the take. Examples: delete the clips shorter than 0.3 s; lower by 3 dB all the clips of a character. Locked tracks are not touched; one undo reverts.



The logical editor.

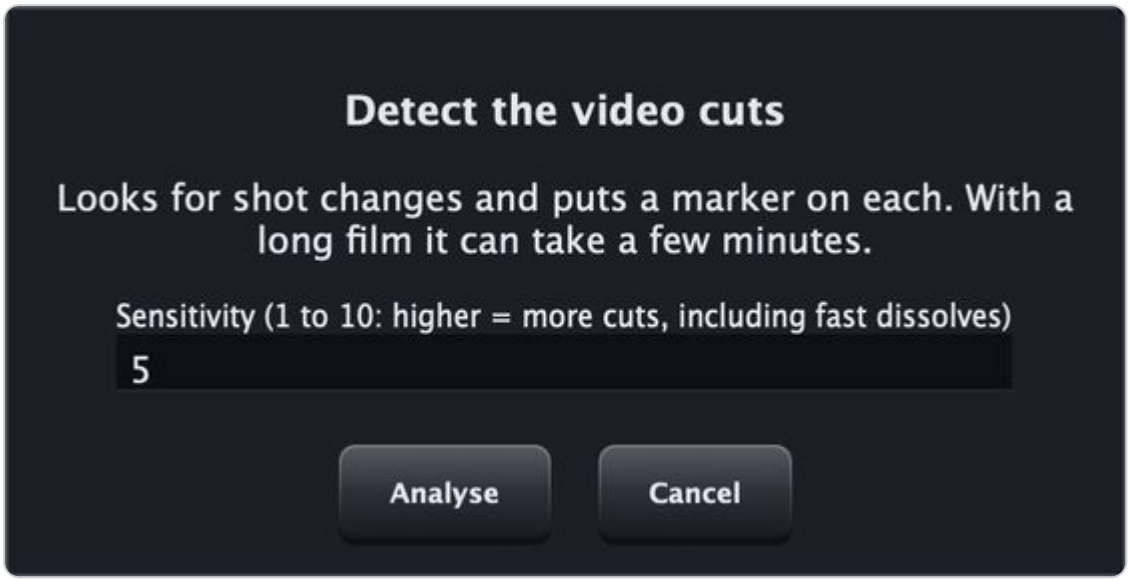
Edit > Batch rename… renames the selected clips or tracks together, or all the markers, with a pattern: `{n}` running number (with a first number and digits of your choice), `{nome}` current name, `{traccia}` (track), `{personaggio}` (character), `{cue}`; *Find / Replace* works on the current name first. A clip's name is just a label: the file is not renamed.



Batch rename.

7.13 Markers

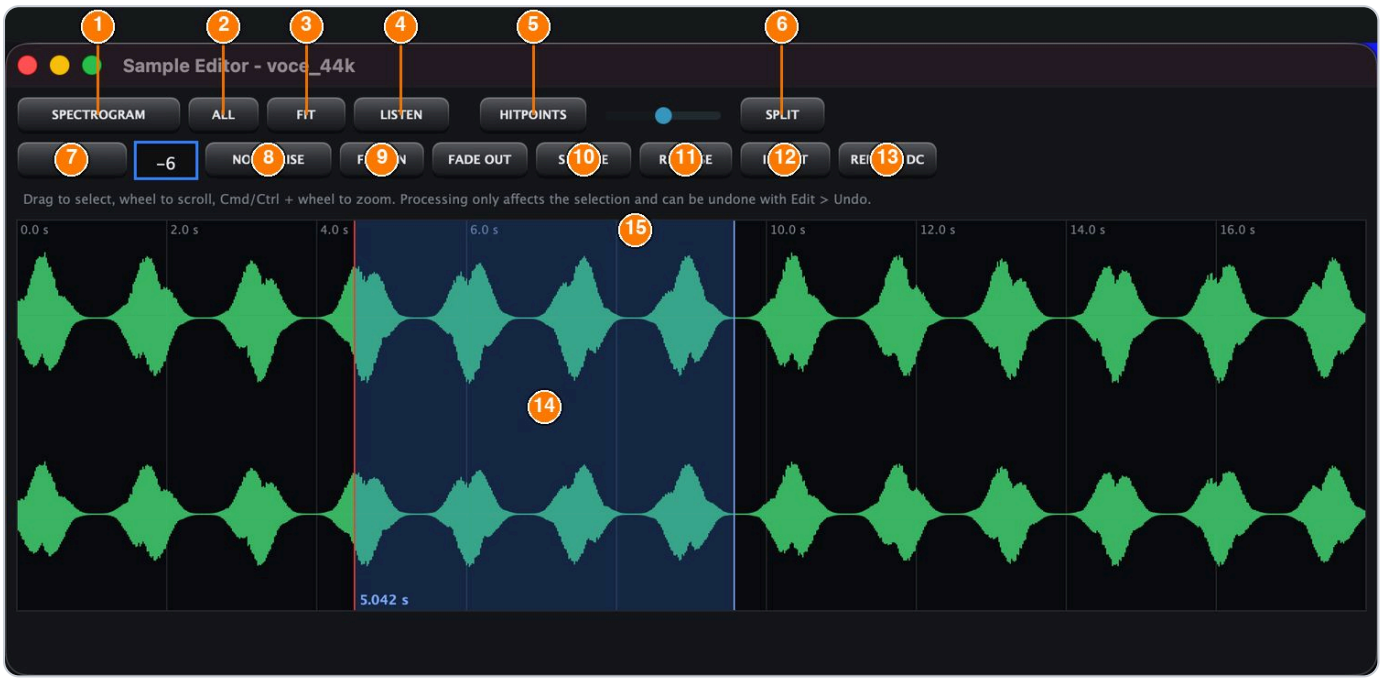
The **Marker** menu adds a marker at the playhead (M), creates a cycle marker from the loop region (with start and end), jumps to the previous or next one (Alt + ←/→), exports the list to CSV and **detects video cuts** putting a marker on every change of shot (sensitivity 1 to 10: higher = more cuts, including fast dissolves). A session's markers appear at the bottom of the menu: a click takes the playhead there.



Detect the video cuts.

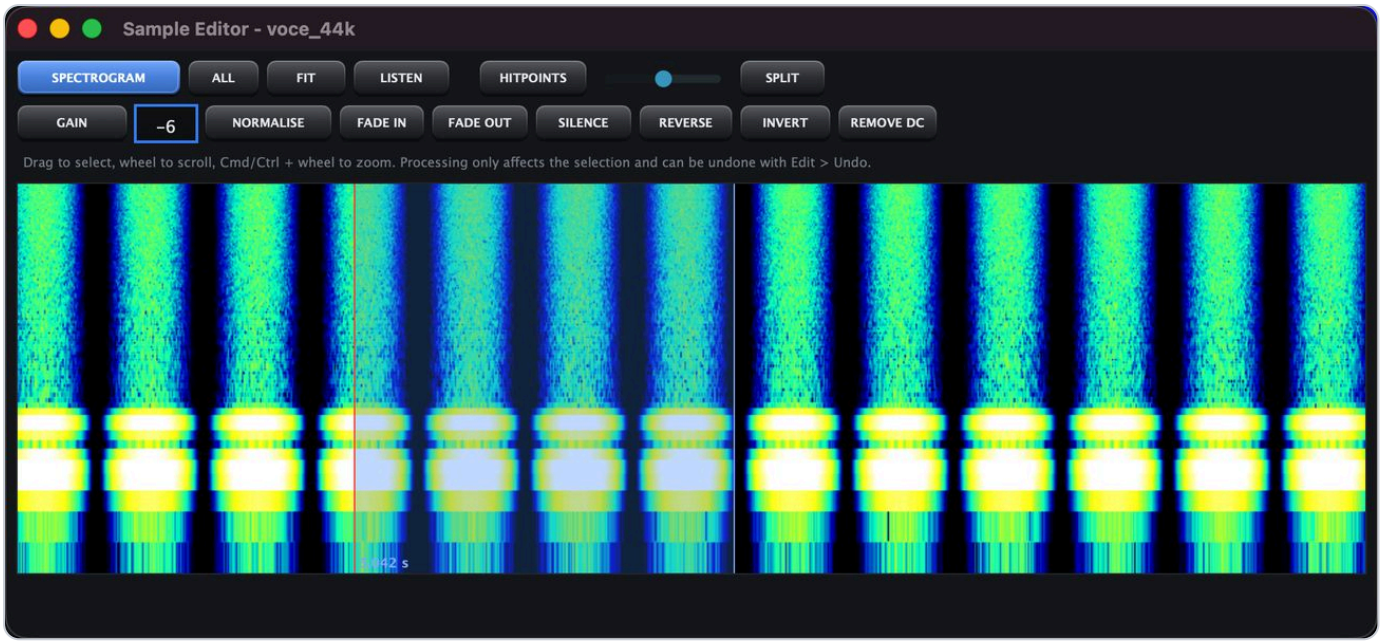
8. The Sample Editor

A **double-click on a clip** (with the Select tool) or **Edit > Sample Editor (selected clip)** opens a window dedicated to that clip, with the waveform at full width, zoom, selection of a portion and processing of the selection only.



The Sample Editor with a portion selected.

1. **SPECTROGRAM**: switches between waveform and spectrogram.
2. **ALL**: selects the whole clip (like a double-click).
3. **FIT**: shows the whole clip.
4. **LISTEN**: plays the selection (or the whole clip) in a loop; stop with the space bar.
5. **HITPOINTS**: shows the hitpoints; the slider next to it sets the sensitivity.
6. **SPLIT**: splits the clip at the hitpoints shown.
7. **GAIN**: changes the selection by the dB written in the box next to it.
8. **NORMALISE**: brings the selection's peak to –1 dBFS.
9. **FADE IN / FADE OUT**: fades the selection in or out.
10. **SILENCE**: silences the selection (cuts out a cough, a click).
11. **REVERSE**: reverses the selection in time.
12. **INVERT**: inverts the selection's polarity.
13. **REMOVE DC**: removes the DC offset from the selection.
14. **Selection**: drag on the waveform; the length appears at the bottom. A click moves the playhead.
15. Ruler in seconds: the wheel scrolls, **Cmd + wheel** (or pinch) zooms.



The spectrogram view: time horizontally, frequency (60 Hz – 12 kHz, logarithmic scale) vertically and level in colour.

- Every processing writes a **new file** (in the Elaborati folder) and replaces the clip's audio without changing its position and length; **Edit > Undo** brings it back. Envelope and sync point follow the new file.
- The spectrogram helps to see sibilance, background noise, wind bursts and clicks that the ear finds hard to locate.

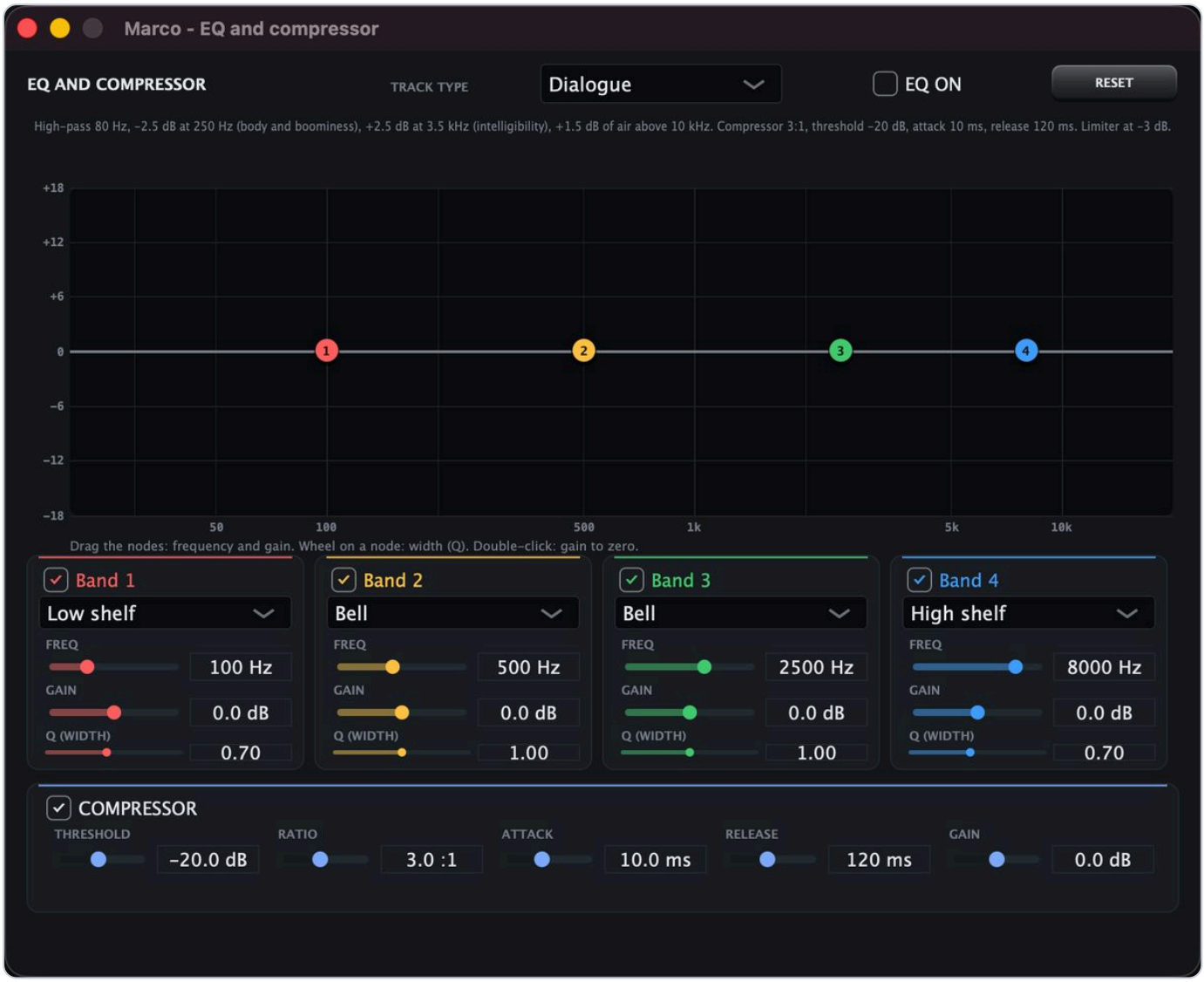
9. Mixer and sound processing

9.1 Volume, pan, mute, solo and sends

- **Fader**: from –∞ to +6 dB; double-click = 0 dB. The **pan** is a knob from L to R. Each track's **input gain** (Trim) is set with the knob in the track header (where there is room).
- **M**, **S**, **Ø**: mute, solo and polarity inversion. With at least one solo active only the soloed tracks play. The **M** button at the top left of the timeline clears all mutes, **S** all solos.
- **SEND A / SEND B**: sends (post-fader) to the two effect returns **FX A** and **FX B**, which have their own inserts, fader and mute. They are for a reverb or echo shared by several tracks.
- **Track output** (MASTER button): the track can go to the master or to one of the four buses (Dialogue, M&E, Effects, Guide); the buses have their own fader and inserts.

9.2 The EQ and compressor editor

A channel's **EQ** button opens the track's editor: a **4-band EQ** with an interactive graph and a **compressor**. The track type (Dialogue, M&E, Effects, Guide) can also be chosen here.



A track's EQ and compressor editor.

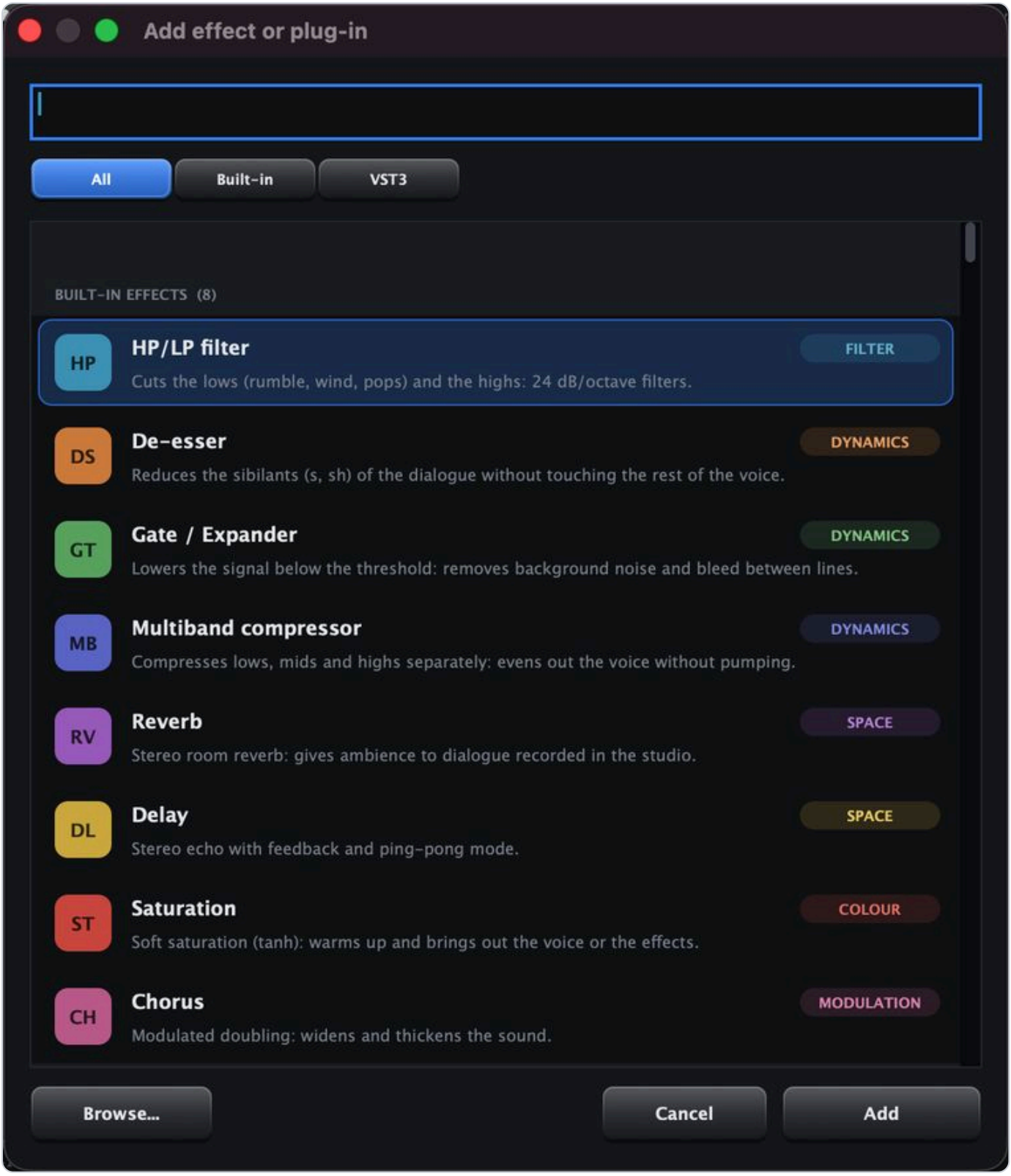
- **Graph**: drag the points to change frequency and gain; the wheel on a point changes the width (Q); double-click resets the gain.
- **Bands 1–4**: type (Bell, Low shelf, High shelf, High-pass, Low-pass), frequency, gain and Q; the checkbox enables or bypasses each band, **EQ ON** the whole equaliser.
- **Compressor**: threshold, ratio, attack, release and make-up gain.

9.3 Track and master limiter

- **LIM** (track): limiter after the fader, ceiling set with the arrow beside it; it adds 3 ms of latency (compensated) and lights up yellow when it acts.
- Master **LIMITER**: 3 ms look-ahead and 80 ms release; no peak exceeds the ceiling (knob, double-click –1 dB).
- **MONO** and **DIM** are monitoring controls: they do not affect export and measurements.

9.4 Inserts: built-in effects and VST3 plug-ins

Each channel (and the master, the FX returns and the buses) has four inserts. Clicking "+ Insert" (or **Track > Add VST3 plug-in to the selected track**, key P) opens the **effect browser**, with a search box for name or category and the tabs *All*, *Built-in* and *VST3*.



The effect browser.

- **Built-in effects** (eight): HP/LP filter at 24 dB/octave, de-esser, gate/expander, multiband compressor, room reverb, stereo delay with ping-pong, saturation, chorus.
- **VST3 plug-ins**: the program scans the standard folders; **Browse…** adds one by hand. Plug-in state is saved with the session. A plug-in that fails to load stays in the list with an error message.
- **Latency compensation (PDC)**: the program aligns the tracks by compensating the plug-ins' latency (**Transport > Plug-in latency compensation**); a bypassed insert does not count.
- Right-clicking an insert: open the editor, bypass, move up/down, remove.

9.5 Automation

Every track has automation curves for **VOL**, **PAN**, **SEND A** and **SEND B**. The AUTO button in the track header chooses the mode (off, read, write) and the button next to it which curve to show. With the **Automation** tool (U) points are added, moved and deleted; in write mode fader moves during playback record the curve. **Clear automation** (Track menu) resets the displayed curve.

9.6 Loudness measurement and spectrum

- The analysis panel measures the master in real time according to **ITU-R BS.1770 / EBU R128**: integrated, momentary, short-term loudness, loudness range and true peak; it chooses the standard (EBU R128 –23 LUFS, ATSC A/85 –24, ARIB TR-B32, Netflix dialogue –27, podcast, streaming, audiobooks) and shows its target and ceiling.
- **SPEECH ONLY** limits the measurement to blocks with speech (estimated from energy): it is the value on which delivery "dialogue gating" is based.
- The **SPECTRUM** tab shows a real-time spectrum (8192-point FFT, 20 Hz – 20 kHz) with held peaks (**PEAKS**) and a +4.5 dB/octave tilt (**TILT**) that makes pink noise look flat.

10. The ADR workflow

10.1 ADR cues

A **cue** is a line to record: it has a number, a **character**, a **scene**, a **reel**, **in** and **out** (timecode), the text of the **line**, **notes** and a **status**: *To record*, *Recorded*, *Approved*, *To redo*. On the timeline the cue is a coloured zone (with the number and character at the top) with handles on its edges.

- **Changing a cue on the timeline**: drag the zone's left or right edge for in and out, the body (the strip at the top) to move it; it snaps to clips, markers and the playhead. Selecting a cue moves the playhead to its in point and selects it in the ADR window (and vice versa).
- **New cue**: *ADR > New cue from the loop region or the playhead* (or the **+ CUE** button): creates a cue from the loop region, or of 4 seconds from the playhead.

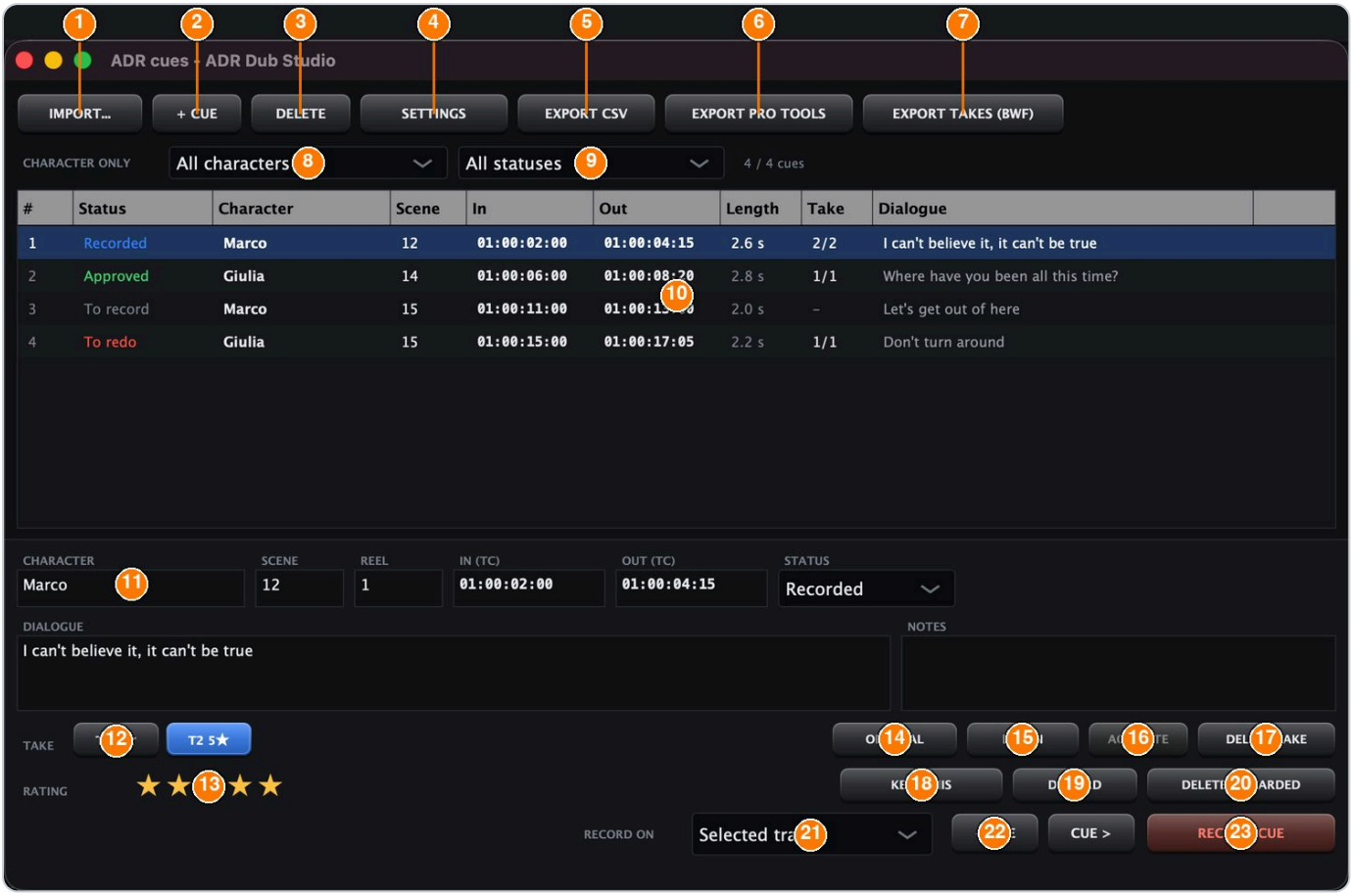
10.2 Importing cues

ADR > Import cues from Pro Tools / EDL / table… (or the **IMPORT…** button of the ADR window, or by dragging the file onto the tracks) opens a dialog that recognises the file format by itself and shows the timecode and the options.

Format	What is read
Pro Tools (Session Info text)	The marker list (Marker Listing): position in absolute timecode or samples, name and comments. In the comments the program reads `CHARACTER:`, `SCENE:`, `REEL:`, `OUT:`, `DIALOGUE:`, `NOTES:`; a free comment becomes the line and the name the character.
EDL CMX3600	Events and locators with comments (also from Nuendo); the frame format is checked (an impossible frame in the chosen format gives a warning).
CSV / TSV table	A timecode column plus the columns the program recognises from the headings (Italian or English): character/actor/role, scene, reel, in/start, out, dialogue/line/text, notes/comment. It works with Nuendo, Fairlight, Resolve and ADR reports.
SRT / WebVTT subtitles	Each subtitle becomes a cue with its line (without a character: you assign it afterwards).

- **Timecode format and start**: the dialog proposes the session's; cues before the session start are discarded with a warning.
- **Default length**: for cues without an out point.
- **Replace the existing cues**, **Also create a marker for each cue** (like Memory Locations) and **Create a Dialogue track for each character** (with role, bus and the "Dialogue" group).

10.3 The ADR window



The ADR window with cue 1 selected.

1. **IMPORT…**: imports cues from a file.
2. **+ CUE**: creates a cue from the loop region or the playhead.
3. **DELETE**: deletes the selected cue.
4. **SETTINGS**: opens the session settings.
5. **EXPORT CSV**: the cue list as CSV (cue sheet).
6. **EXPORT PRO TOOLS**: the cues as Pro Tools marker text.
7. **EXPORT TAKES (BWF)**: the active takes as BWF files with timecode.
8. Filter by **character**: the list and the previous/next CUE buttons follow the filter.
9. Filter by **status** (for example "To record").
10. **Cue list**: status (colour), character, scene, in, out, length, takes (active/total) and line. Double-click a row to move the playhead to the cue.
11. Editable **details** of the cue: character, scene, reel, in and out (TC), status, line and notes.
12. **Takes** (T1, T2…) with their stars: blue is the active one.
13. **Rating** 1–5 stars of the chosen take (a second click on the same star removes it).
14. **ORIGINAL**: listen to the original line (production audio, without the ADR takes) with the video in sync.
15. **LISTEN**: listen to the chosen take alone, with the video in sync.

16. **ACTIVATE**: makes the chosen take active: it is the one that plays in the mix and is exported.
17. **DELETE TAKE**: deletes the chosen take (undoable).
18. **KEEP THIS**: keeps the chosen take (makes it active) and discards all the others of the cue.
19. **DISCARD**: discards the take (it no longer plays or counts) or restores it; it does not delete it from disk.
20. **DELETE DISCARDED**: permanently deletes the cue's discarded takes (with Shift, of all cues).
21. **RECORD ON**: the track to record on (normally the character's Dialogue track).
22. **< CUE / CUE >**: previous and next cue (Cmd + Shift + ←/→).
23. **RECORD CUE**: pre-roll, beeps, streamer over the picture and automatic punch on the selected cue.

10.4 Recording a cue

1. In *Session settings* set pre-roll, post-roll and number of beeps; choose the input in *File > Audio settings*.
2. In the ADR window select the cue and, in **RECORD ON**, the destination track.
3. Press **RECORD CUE**. The program moves the playhead to the pre-roll, shows the video (if it was hidden) and starts: **countdown** with the beeps (in the director's headphones and at the output), an **amber bar** with the big number of seconds over the picture, then a **red bar** and a flashing red frame for the length of the cue, with the line as an overlay.
4. The recording is an **automatic punch**: it writes only between the cue's in and out points. At the end a **new take** (T1, T2…) is created on that track, compensating for the interface's latency, and the cue becomes *Recorded*. Repeat as often as you like: takes overlap and only the active one plays.
5. **Stopping**: *ADR > Stops the recording of the cue* (Cmd + Shift + R) or the stop button.

> **Note** The beeps are always heard, even with DIM on: they are added after the monitoring controls. If the cue is very close to the start and there is no room for the whole pre-roll, the beeps get closer together instead of disappearing.

10.5 Choosing the best take

- Select the cue and choose a take with the **T1, T2…** buttons (the star next to it is the rating). **LISTEN** plays it alone; **ORIGINAL** compares it with the original line.
- Give **stars**; at the end **KEEP THIS** makes the best one active and discards the others; **DISCARD** hides a take without deleting it; **DELETE DISCARDED** deletes.
- The cue's status (*Recorded*, *Approved*, *To redo*) is changed in the details; the **Session check** reports cues still to record, to redo, or with several takes to choose from.
- In the timeline inactive takes are grey; right-clicking an ADR clip offers *activate the take*, *delete the take* and *fit to the cue's length (time-stretch)*.

10.6 Importing takes already recorded

ADR > Import recorded takes from a folder… reads a folder of files named like `Cue012_Marco_Take2_BEST.wav`: it matches each file to the cue by number, checks the character, recognises

the take number and the word "BEST" (the chosen take) and places the file at the timecode written in the BWF (otherwise at the cue's in point). Files already present are not duplicated and anomalies are reported in a list.

10.7 Exporting cues and takes

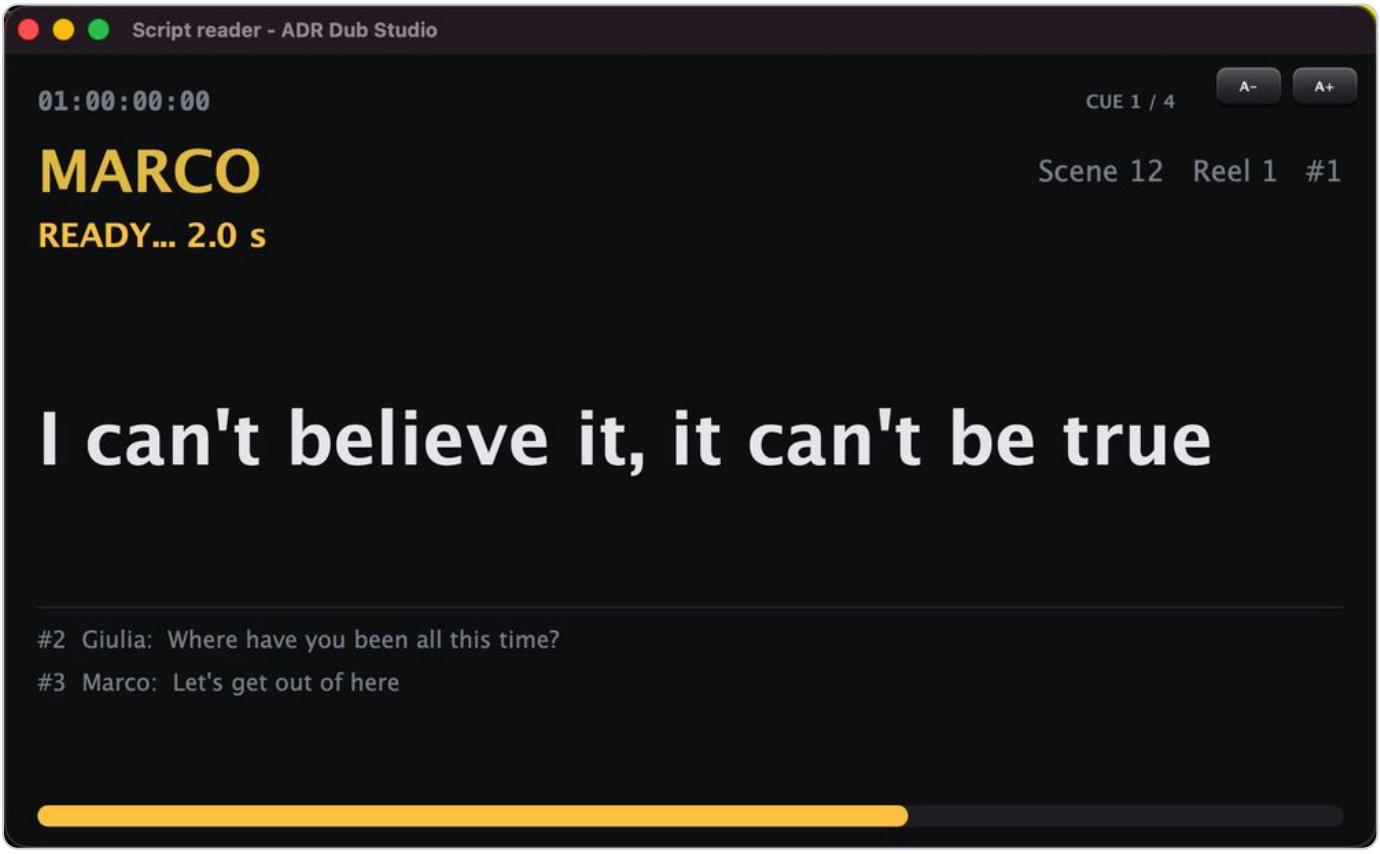
- **Export cue sheet (CSV)** and **Export cues as Pro Tools markers (text)**: they read back with the same cues.
- **Export active takes (BWF with timecode)**: one 24-bit BWF file for every cue with an active take, with the timecode (time reference) of the position on the timeline, ready to place in Pro Tools.
- **Cue sheet and script per character (printable)**: an HTML page with the summary of all cues and one script per character; it opens in the browser, from where it can be printed or saved as PDF.

10.8 The recording room

Three tools designed for the dubbing room, with the actor inside and the director in the booth.

SCRIPT READER FOR THE ACTOR

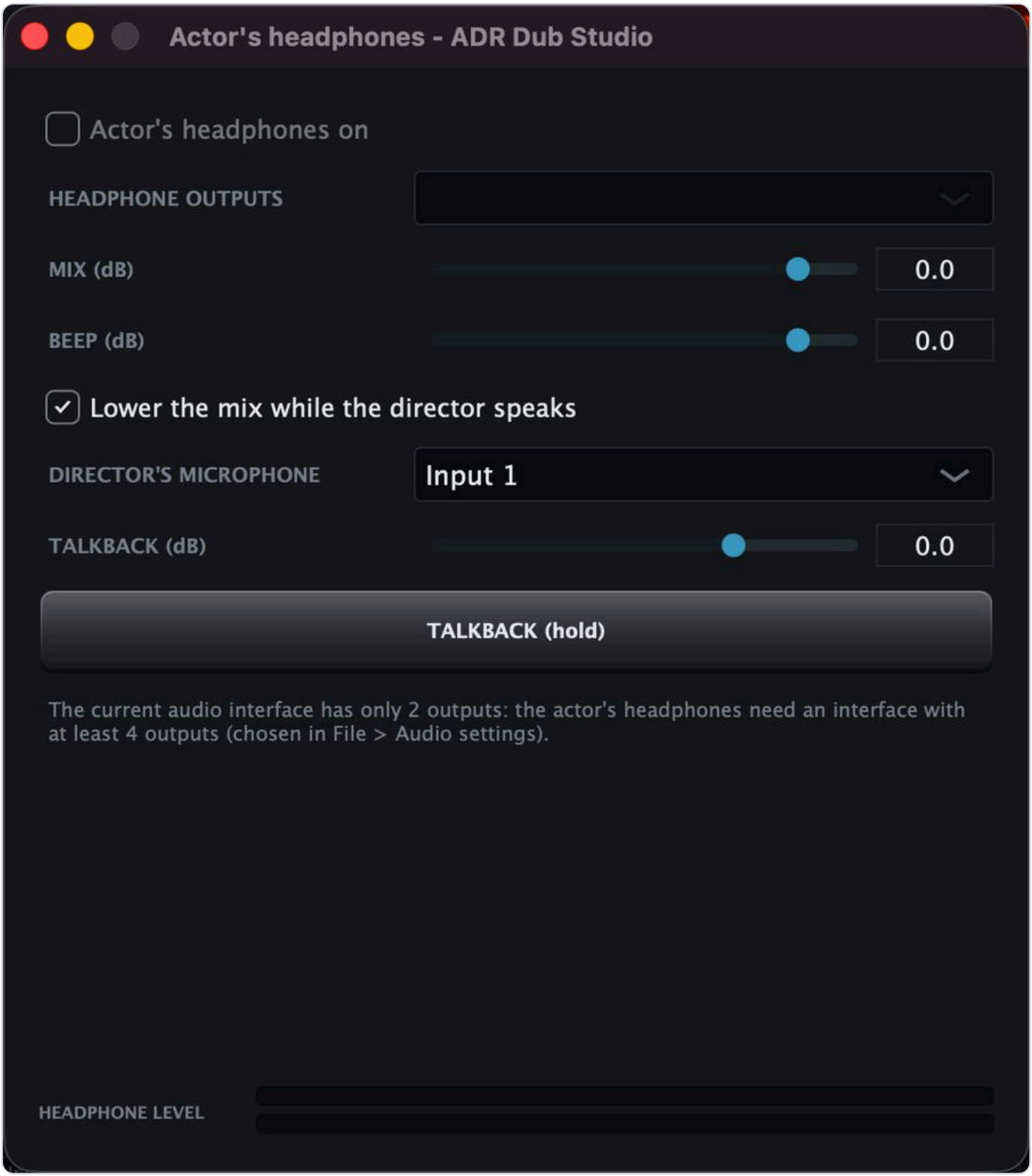
ADR > Script reader for the actor opens a window (to be moved to the actor's screen) with the character, scene and reel, the timecode, the **line in large type** with a "READY…" indication before the in point, a progress bar and the next two lines. **A– / A+** (or the wheel with Cmd) change the text size.



The script reader for the actor.

ACTOR'S HEADPHONES AND TALKBACK

ADR > Actor's headphones and talkback sends a mix for the actor's headphones to a **second pair of outputs** of the interface (an interface with at least 4 outputs is needed), independent from the speakers' master.



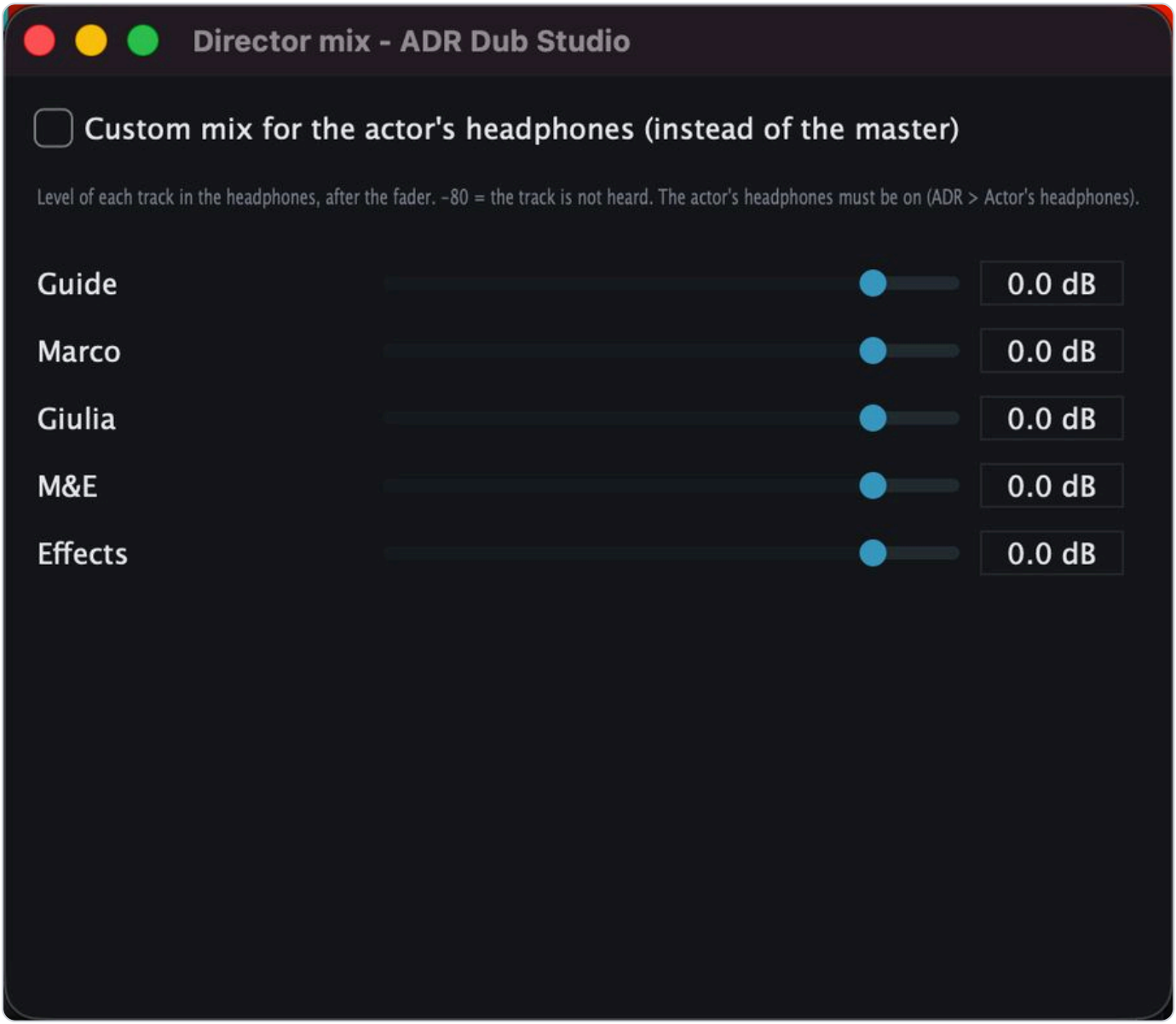
Actor's headphones and talkback.

- **Actor's headphones on** and **Headphone outputs** (the output pair).
- **MIX**, **BEEP**, **TALKBACK** (dB): levels of the mix, the beeps and the director's voice in the headphones.

- **Lower the mix while the director speaks**: during talkback the mix goes down.
- **Director's microphone**: the microphone input; **TALKBACK (hold)**: talk into the actor's headphones while pressed.
- The **HEADPHONE LEVEL** meter at the bottom shows the signal being sent.

DIRECTOR MIX

Normally the actor hears the master. With **ADR > Director mix for the headphones (levels per track)** you can instead give them a mix of their own: *Custom mix for the actor's headphones (instead of the master)* and, for each track, the level in the headphones after the fader (–80 = not heard). For example the guide and the music loud and with no effects. The custom mix is saved with the session and only works with the actor's headphones on.

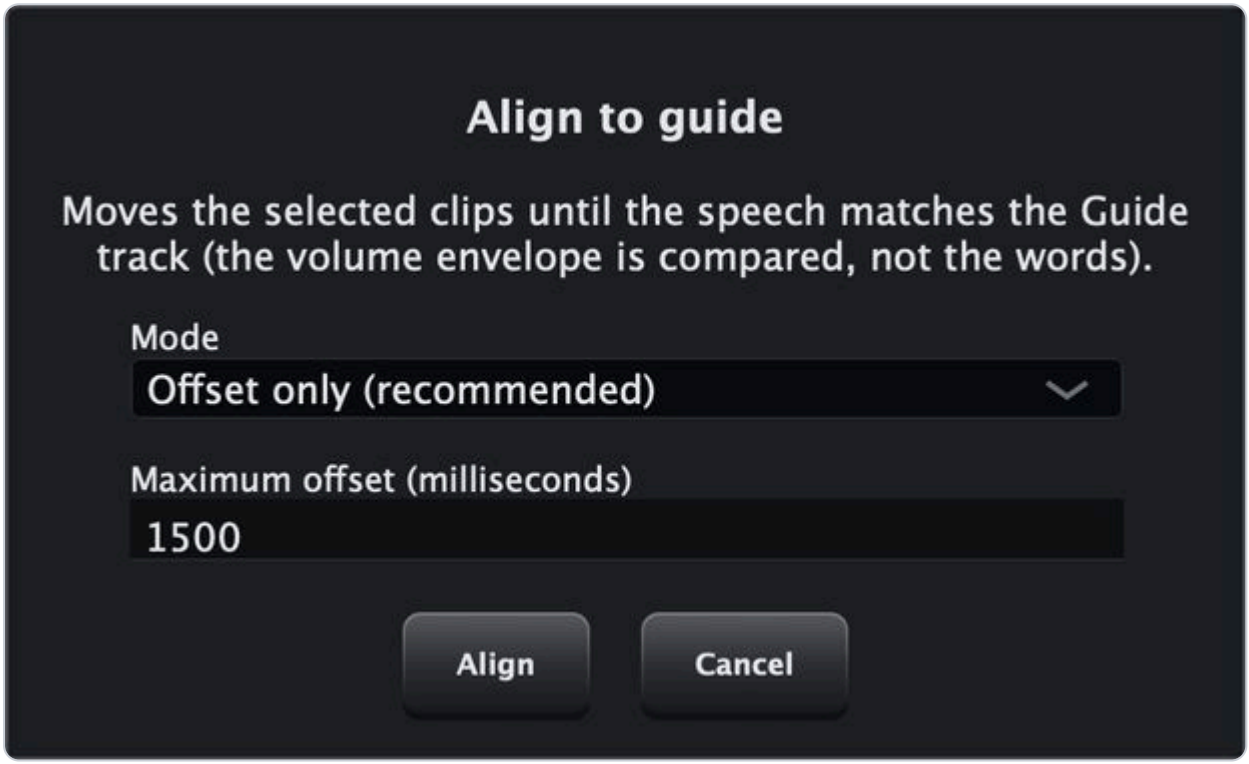


The director mix for the headphones.

11. Sync and conforming

11.1 Align to guide

Edit > Align to guide (automatic sync) moves the selected clips (or the selected cue's active take) until their speech coincides with that of the **Guide** track. The program compares the volume envelope (not the words) and searches up to the **maximum offset** you set (default 1500 ms); an unreliable result (similarity below 0.55) is not applied and is reported. With *Offset and adapt the speech duration* it also adjusts the duration (stretch) between 80 % and 125 %.



Align to guide.

11.2 Align phrase by phrase

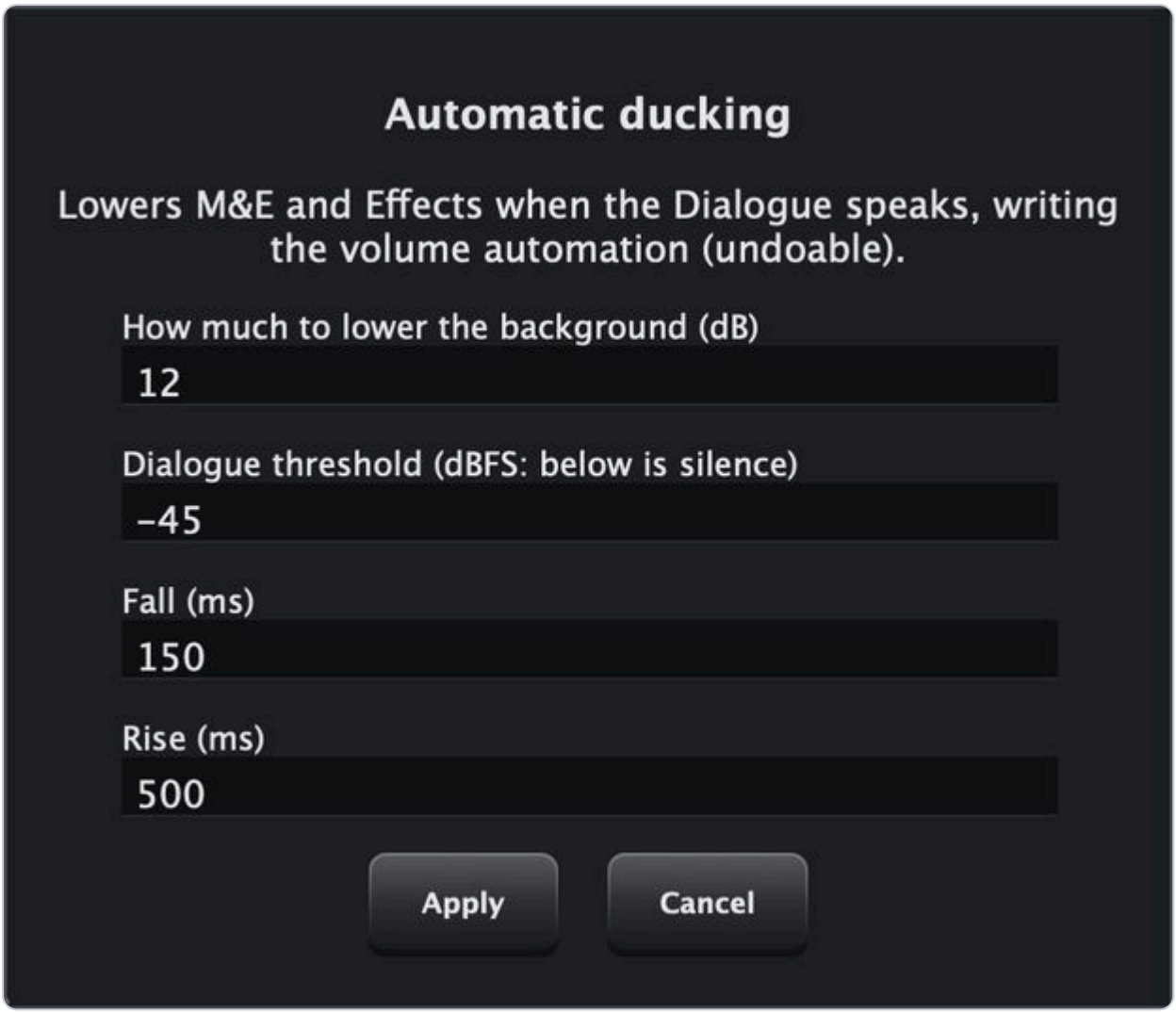
Edit > Align to guide phrase by phrase (Time Warp) goes further: after the global shift it identifies the take's **phrases** (zones with speech, separated by pauses) and matches them to the guide's phrases; then with a Time Warp it brings the start of each phrase onto that of the matching phrase, and the end too if the length is similar (between 85 % and 120 %). The pauses between phrases get shorter or longer; pitch does not change. The report says how many phrases out of how many were matched. One undo per clip.



Align phrase by phrase.

11.3 Automatic ducking

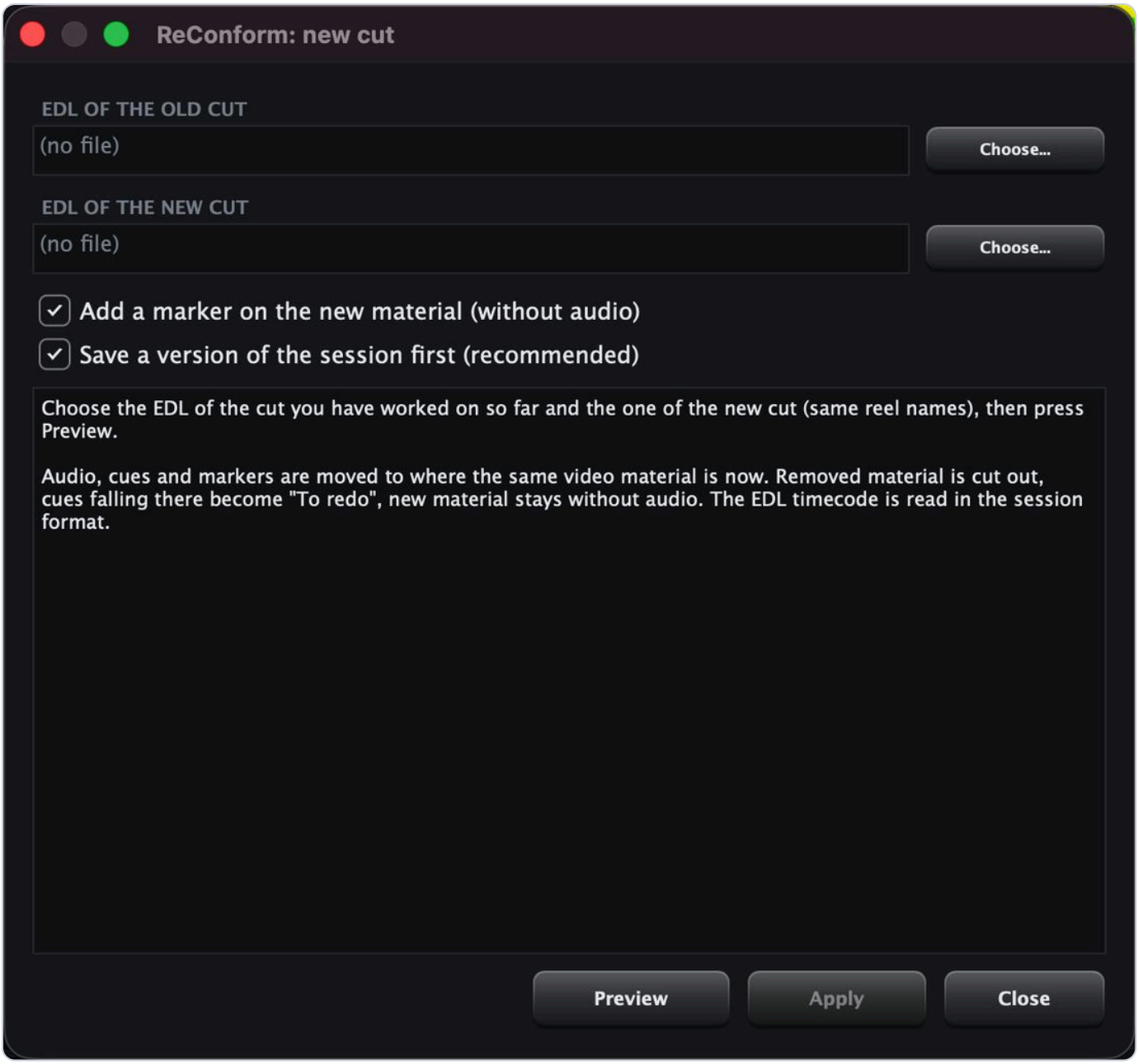
Track > Automatic ducking of the background under the dialogue lowers **M&E and Effects** when the **Dialogue** speaks, writing the volume automation (undoable). You choose how much to lower (dB), the dialogue threshold, and the **fall** and **rise** times.



Automatic ducking.

11.4 ReConform: a new cut

When a new cut of the film arrives, **Edit > ReConform** moves audio, cues and markers to their new positions starting from the **EDLs** (CMX3600) of the old and new cut, with the same reel names. Removed material is cut, cues that fall in the removed part become "To redo" with the note "Outside the new cut", new material stays without audio; markers can be put on the edit points. A text **Preview** first lists the shifted segments, the seconds removed and inserted, the cues touched and the warnings; **Apply** does everything in one undo step (first saving a safety version, if you ask). The new video is imported separately.

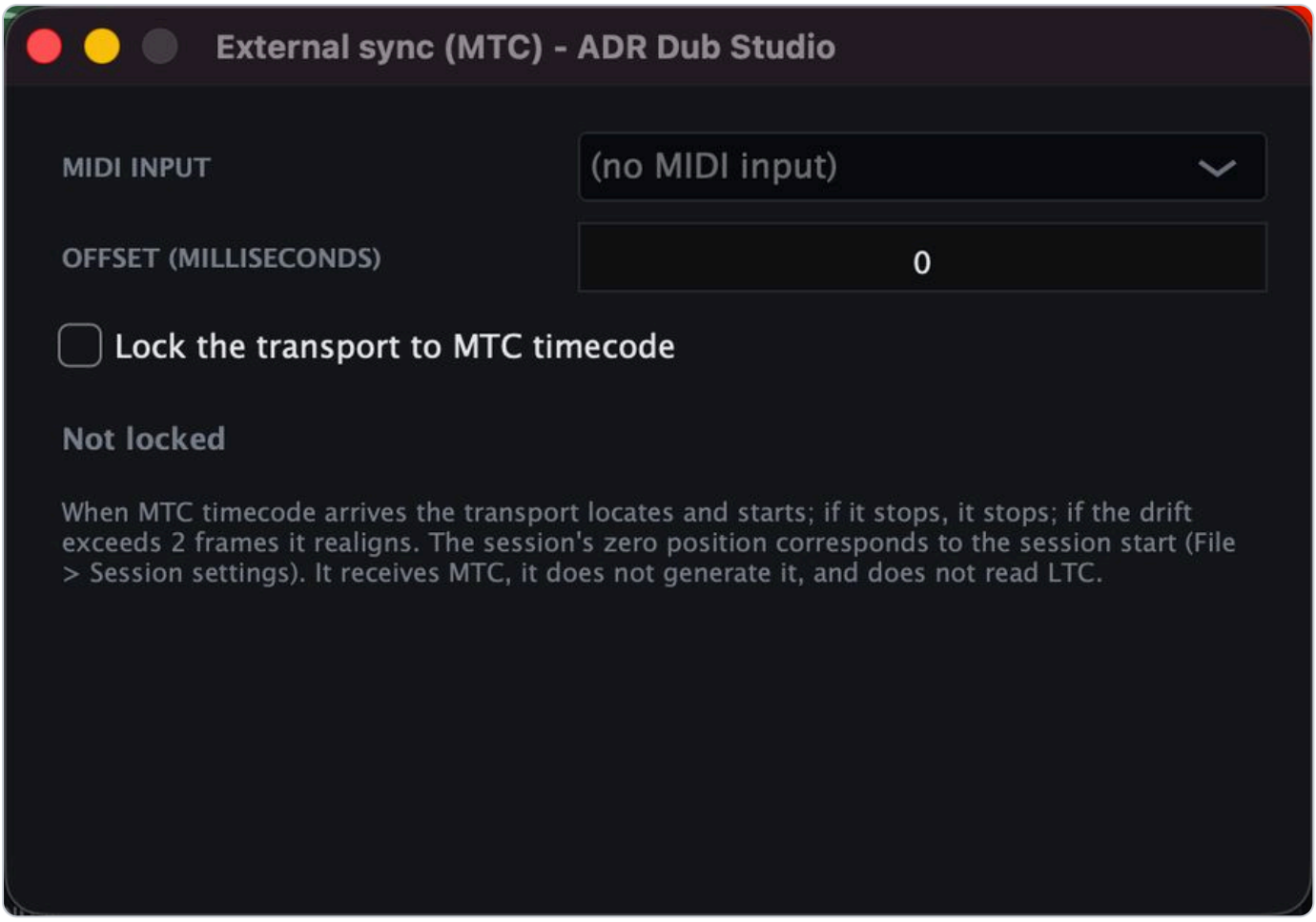


ReConform.

> **Note** EDLs with dissolves or wipes are treated as cuts at mid-transition; speed remaps are not supported.

11.5 External sync MTC

Transport > External sync (MTC) makes the transport follow an external **MIDI timecode (MTC)**: when timecode arrives the program locates and starts; if it stops, it stops; if the drift exceeds 2 frames it realigns. You choose the MIDI input and an offset in milliseconds; the session's zero position corresponds to the session start. It receives MTC: it does not generate it and does not read LTC.

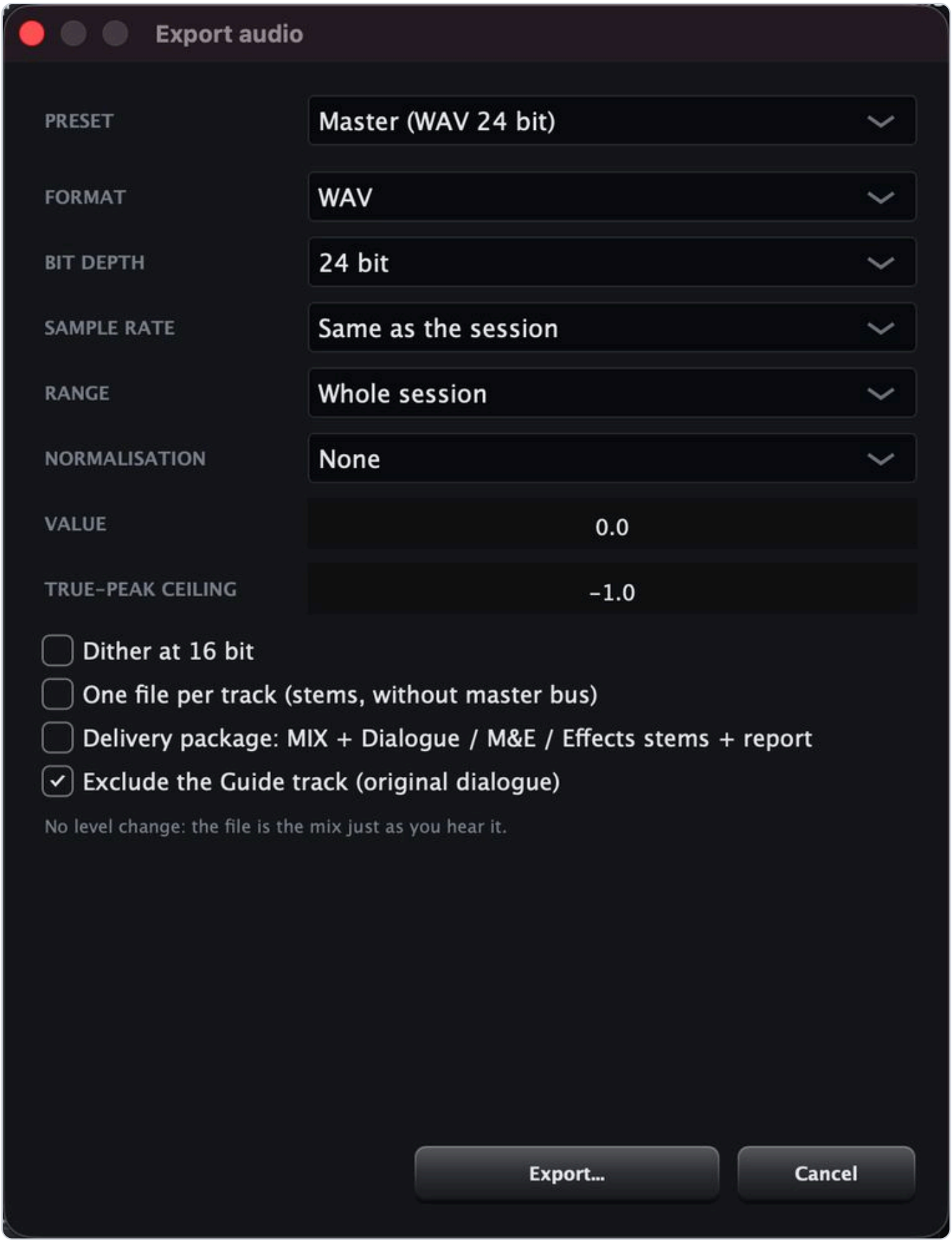


External sync MTC.

12. Export and delivery

12.1 Exporting the mix

File > Export audio (Cmd + E) writes the mix (or the stems) to a file. The export is offline (faster than real time) and uses the same inserts, the PDC and the master limiter.

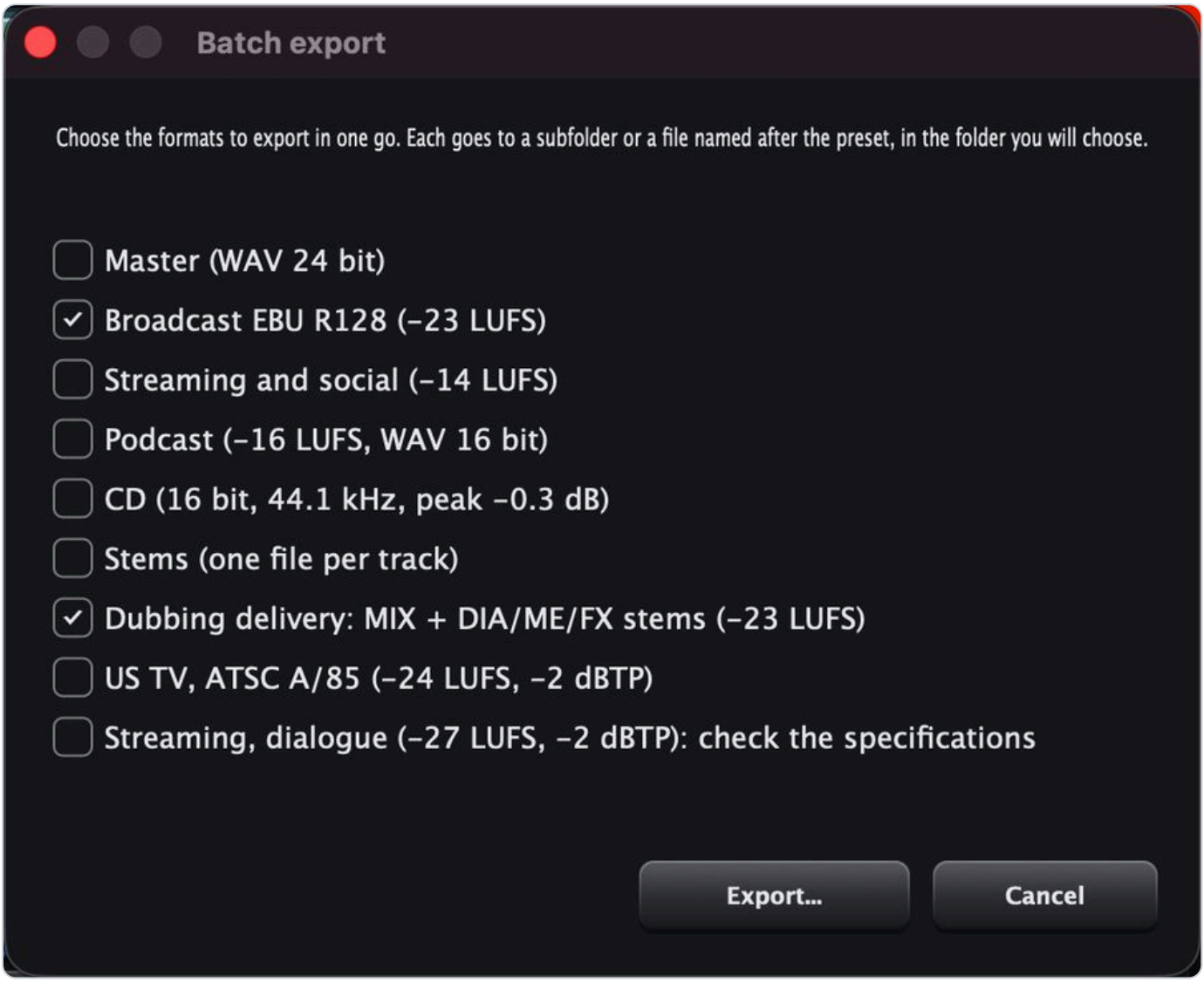


File > Export audio.

Item	What it does
Preset	Master WAV 24 bit, Broadcast EBU R128 (–23 LUFS), Streaming and social (–14 LUFS), Podcast (–16 LUFS, WAV 16 bit), CD (16 bit, 44.1 kHz, peak –0.3 dB), Stems, Dubbing delivery, US TV ATSC A/85 (–24 LUFS, –2 dBTP), Streaming dialogue (–27 LUFS, –2 dBTP).
Format / Bit depth / Sample rate	WAV, AIFF or FLAC; 16, 24 or 32 bit; the session rate, 44.1, 48, 88.2 or 96 kHz.
Range	The whole session, or the range chosen on the ruler (export bar).
Normalisation	None; to peak (value in dBFS); or to **loudness** (target in LUFS): the program measures and brings the programme to the target; if it exceeded the true-peak ceiling a transparent limiter steps in.
Dither at 16 bit	A light dither against quantisation distortion.
One file per track (stems)	One file for every track with clips, after fader and inserts, without the master limiter and plug-ins; the files are aligned with each other.
Delivery package	In one export: the MIX (with the master bus) and one stem for every role bus (Dialogue, M&E, Effects), all with the same gain, plus an **HTML report** with the peak, true-peak and loudness of every file.
Exclude the Guide track	The Guide track (original dialogue) is not included in the export.

12.2 Batch export

File > Batch export exports several formats in one go: tick the presets (for example Broadcast, Streaming, Stems, Dubbing delivery, US TV) and each goes into a subfolder, or into a file named after the preset, in the folder you choose.



Batch export.

12.3 Packages and archives

- **Export the session package (Clip Package)** (File): copies into a folder the session and all the files (with relative paths) to pass to a colleague; it reopens on another computer without relinking anything.
- **Export video with mix** (Shift + Cmd + E): writes a review video with the mix audio in place of the original audio; with the timecode and line overlay (burn-in) (on macOS).
- **Track archive**: see *Tracks*.

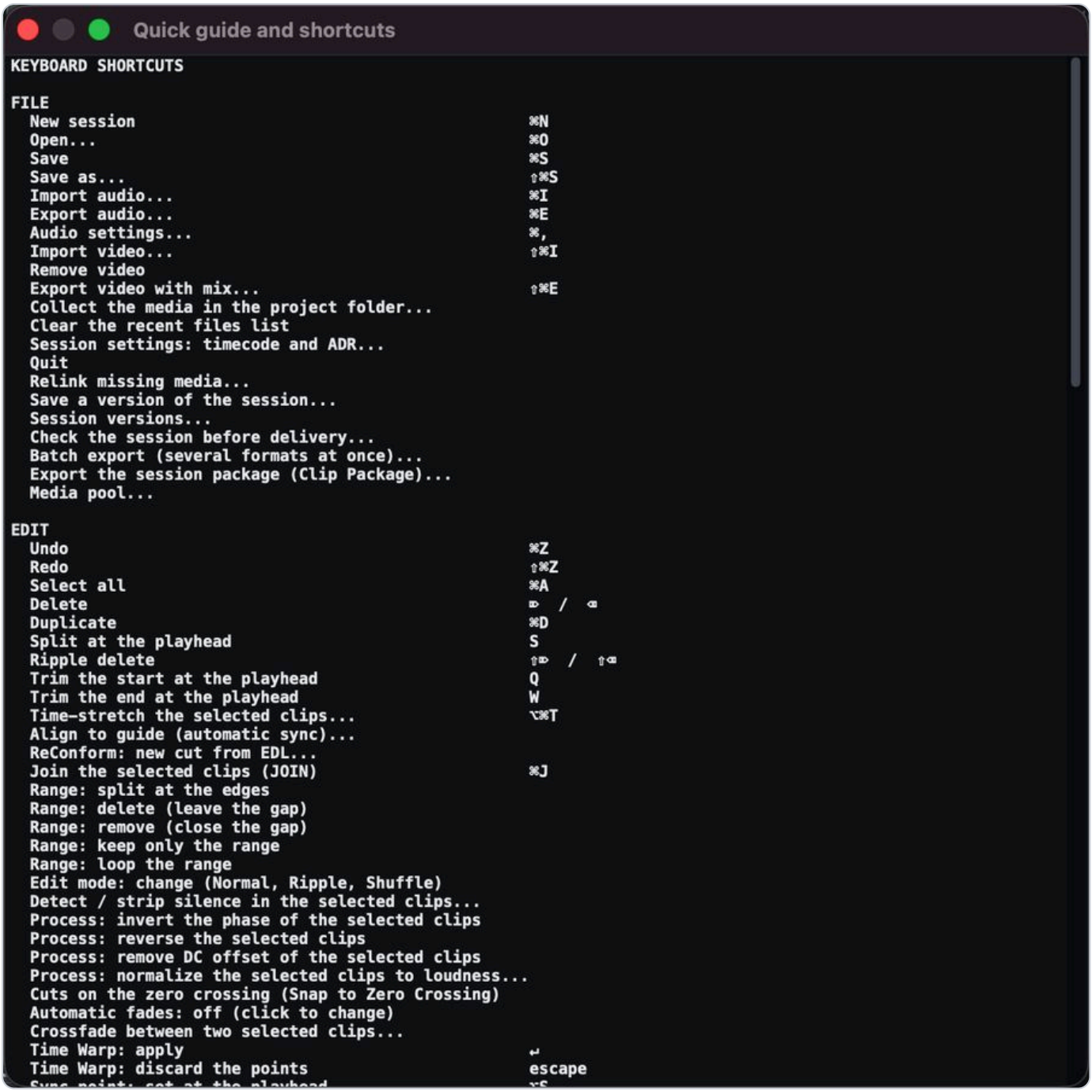
> **Tip** Before delivering run **File > Check the session**: it reports tracks without a role (which would appear only in the MIX), cues to record or redo, missing media.

13. Preferences and customisation

- **Language**: Help > Lingua / Language (Italian, English); it applies on restart.
- **Theme**: View > Theme: Dark, Light, High contrast, Default 5.
- **Layout**: View > Layout: Compact, Standard, Large (width of headers and channels and track height).
- **Track controls**: right-clicking the top-left corner of the timeline chooses which controls to show in the headers (lock, record, mute, solo, trim, meter, automation, input).
- **Keyboard shortcuts**: *Help > Customise keyboard shortcuts* opens an editor in which each command gets the key you prefer (conflicts are flagged; *Restore the defaults* goes back to the original). *Help > Quick guide and shortcuts* (key ?) lists the active ones.
- **Video**: View > Show video / Detach the video window; the detached video can be moved to another screen and re-docked with Shift + D.



The keyboard shortcut editor.



The quick guide with the list of shortcuts.

14. Keyboard shortcuts

Complete list of the default shortcuts (**Cmd** on macOS corresponds to **Ctrl** on Windows). They can be changed from *Help > Customise keyboard shortcuts*.

FILE

Command	Shortcut
New session	`Cmd/Ctrl+N`
Open...	`Cmd/Ctrl+O`
Save	`Cmd/Ctrl+S`
Save as...	`Cmd/Ctrl+Shift+S`
Import audio...	`Cmd/Ctrl+I`
Export audio...	`Cmd/Ctrl+E`
Audio settings...	`Cmd/Ctrl+,`
Import video...	`Cmd/Ctrl+Shift+I`
Export video with mix...	`Cmd/Ctrl+Shift+E`
Quit	`Cmd/Ctrl+Q`

EDIT

Command	Shortcut
Undo	`Cmd/Ctrl+Z`
Redo	`Cmd/Ctrl+Shift+Z`
Select all	`Cmd/Ctrl+A`
Duplicate	`Cmd/Ctrl+D`
Delete	`Delete` / `Backspace`
Split at the playhead	`S`
Join the selected clips (JOIN)	`Cmd/Ctrl+J`
Time-stretch the selected clips...	`Cmd/Ctrl+Alt+T`
Time Warp: apply	`Return`
Time Warp: discard the points	`Esc`
Sync point: set at the playhead	`Alt+S`
Sync point: move onto the cue (or the playhead)	`Shift+Alt+S`
Ripple delete	`Shift+Delete` / `Shift+Backspace`
Trim the start at the playhead	`Q`
Trim the end at the playhead	`W`

TRACK

Command	Shortcut
New track	`Cmd/Ctrl+Shift+T`
Arm/disarm selected track	`A`
A/B comparison with the reference	`` ` ``
Move the track up	`Cmd/Ctrl+Alt+↑`
Duplicate the track	`Cmd/Ctrl+Shift+D`
Add VST3 plug-in to the selected track...	`P`

TRANSPORT

Command	Shortcut
Play / Stop	`Space`
Go to start	`Home`
Go to end	`End`
Loop	`L`
Loop from selection	`Shift+L`
Back one frame	`←`
Forward one frame	`→`
Back one second	`Shift+←`
Forward one second	`Shift+→`
Stop recording	`R`
Monitor the input while recording	`I`

VIEW

Command	Shortcut
Snap	`N`
Follow the playhead	`F`
Zoom in	`=` / `+`
Zoom out	`-`
Fit to screen	`Shift+Z`
Show mixer	`Shift+M`
Show video	`Shift+V`
Clip info (Info Line)	`Cmd/Ctrl+Alt+I`
Dock the video window	`Shift+D`
Fit track height	`Shift+F`
Advance the video by 1 ms	`,`
Delay the video by 1 ms	`.`
Taller tracks	`Cmd/Ctrl+↑`
Shorter tracks	`Cmd/Ctrl+↓`

TOOLS

Command	Shortcut
Selection	`V`
Razor	`B`
Stretch	`E`
Slip	`Y`
Range	`X`
Mute	`K`
Eraser	`G`
Comp	`C`
Hand	`H`
Zoom	`Z`
Time Warp	`O`
Clip envelope	`D`
Scrub	`J`
Automation	`U`

ADR

Command	Shortcut
ADR window	`Cmd/Ctrl+Shift+A`
Stops the recording of the cue	`Cmd/Ctrl+Shift+R`
Next cue	`Cmd/Ctrl+Shift+→`
Previous cue	`Cmd/Ctrl+Shift+←`

HELP

Command	Shortcut
Quick guide and shortcuts...	`?`

MARKER

Command	Shortcut
Add marker at the playhead	`M`
Previous marker	`Alt+←`
Next marker	`Alt+→`

15. FAQ and troubleshooting

Problem	Solution
I hear nothing / playback does not start	Check the output in File > Audio settings and the **M** button of the master and tracks; with a solo active only soloed tracks play. If the audio interface was unplugged, choose the output again.
I do not hear the beep or see the countdown when recording	In Session settings set the beeps (at least 1) and the visual streamer; the count appears over the picture: if the video is hidden the program shows it by itself. The level of the beeps in the headphones is set in *Actor's headphones*.
"I cannot open an audio input"	Allow the microphone in System Settings > Privacy & Security > Microphone and choose the input in File > Audio settings.
Video and audio are out of sync	Adjust the video offset under the viewer (–F/+F, –/+, or typing the milliseconds). For ADR takes use *Align to guide* or the sync point.
A clip is hatched with the text OFFLINE	The audio file was not found: use File > Relink missing media and choose the folder that contains it.
I cannot edit a clip	The track is locked (**L** button lit) or frozen: unlock or unfreeze it from the track menu.
The cue import finds nothing	Check the timecode format and the session start in the import dialog: cues before the session start are discarded. A CSV table needs a timecode column.
The program closed suddenly	On reopening accept to **restore** the safety copy. Recordings are always on disk in the media folder.
With a 2-output interface the actor's headphones do not work	At least 4 outputs are needed: the second pair feeds the headphones.

15.1 Known limits

- Time Warp, time-stretch and phrase alignment change the duration without changing the pitch but, with very strong changes, transients can lose sharpness (it is a WSOLA algorithm suited to voice and dialogue).
- Automatic alignment compares the volume envelope, not the words: with a very noisy guide the result may be uncertain and is reported.
- Track presets do not save plug-ins; mixer snapshots attach to tracks by position (if you reorder tracks, check).
- Surround and Atmos, AAF/OMF/MXF and LTC are not supported; external sync only receives MTC.
- Changing track visibility (hide / show) is not undone by Undo.

15.2 About

ADR Dub Studio is free software released under the **GNU AGPLv3** licence. Copyright © 2026 Fioravante Capaccio. The source code and downloadable versions are on GitHub: `github.com/FioreCapaccio/adr-dub-studio`. Reports and suggestions are welcome.